

The Last Keeper

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Book One — The Last Keeper

CHAPTER TLK-1

The Keeper

On the lighthouse keeper, the automated beacon, and the night the light went out

The Keeper

On a dead man and a burning light, the vigil that could not be automated, and the question that connects a rock in the Irish Sea to every machine you will ever trust with your life.

Smalls Rock sits in the Irish Sea, roughly twenty miles off the coast of Pembrokeshire, Wales. It is barely a rock at all — a low, jagged outcropping that disappears beneath the waves at high tide, invisible to any ship that does not already know it is there. For centuries, it killed sailors. Ships struck the rock in darkness, in fog, in storms, and went down with all hands. The sea around Smalls was a graveyard before anyone thought to put a light on it.

In 1776, a lighthouse was built on the rock — an extraordinary feat of engineering for the era, a wooden structure on iron stilts driven into the stone, exposed to the full violence of the Atlantic. The lighthouse required two keepers at all times. They lived on the rock, maintained the light, and endured conditions that most people today would find difficult to imagine: weeks of isolation, storms that shook the structure to its foundations, waves that broke over the lantern room seventy feet above the waterline.

The keepers were not comfortable. They were not well paid. What they were was necessary.

Without them, the light went dark. And when the light went dark, people died.

In the winter of 1801, one of the two keepers at Smalls — Thomas Griffith — died. The circumstances of his death are not precisely recorded, but what followed is.

His partner, Thomas Howell, was alone on the rock with the body.

A storm was raging. No boat could reach the lighthouse. Howell could not bury the body at sea, because he feared that if the body disappeared, he would be accused of murder when he was eventually relieved.

So he built a crude coffin from salvaged materials and lashed it to the exterior railing of the lighthouse.

The storm persisted for weeks. The coffin broke apart. The body, exposed to the wind and salt, decomposed in view of the lantern room. And Thomas Howell kept the light burning.

He kept it burning for the entire duration of his isolation — some accounts say weeks, others say months, until a relief crew could finally reach the rock.

When they arrived, Howell was emaciated, traumatized, and, according to contemporary reports, so changed by the ordeal that his friends did not recognize him.

But the light had not gone dark. Not for a single night.

Because Howell understood something that no mechanism could understand: the light was not for him. The light was for the ships. And the ships could not know whether the keeper was suffering, or sane, or alone with a dead man in a storm. They could only know whether the light was on.

“Thomas Howell kept the light burning for weeks alongside a decomposing body in an Atlantic storm, because he understood something no mechanism could: the light was not for him. The light was for the ships.”

What a Keeper Does

I begin with this story not because it is pleasant but because it is precise. It isolates, with a clarity that more complex examples cannot match, the thing that is lost when the human is removed from a system. A lighthouse mechanism can keep a flame burning. A clockwork assembly can rotate a lens. A timer can switch a light on at dusk and off at dawn. These are tasks that machines perform more reliably than humans — they do not fall asleep, do not forget, do not get distracted.

On the average night, the machine is better than the keeper.

But Thomas Howell did not do what a machine does. He did not merely maintain a flame. He made a judgment — a continuous, sustained, morally weighted judgment — that the light must remain on regardless of his own circumstances. He evaluated a situation that no designer could have anticipated (a dead partner, a broken coffin, a weeks-long storm), and he chose to keep the light burning despite every human reason to abandon the post. A mechanism does not make this choice, because a mechanism

does not encounter situations it was not designed for.

It encounters the conditions it was built to handle, or it fails. There is nothing in between.

The keeper occupies the in-between — the layer of judgment that operates in the space between what the system was designed for and what actually happens.

The system handles the expected. The keeper handles the rest. And “the rest” is where people live and die.

But the keeper is not always a person in a tower. The judgment that fills the space between design and reality takes four forms, and the chapters that follow remove each of them in turn.

The human keeper: a person exercising situated judgment at the point of consequence — the pilot, the physician, the judge, the watch officer who can see what the system cannot.

The institutional keeper: a procedure, a review board, a regulator, a standards body — the organizational layer that decides what may be built and deployed before it ever reaches a human hand.

The mechanical keeper: a physical interlock, a redundant control, a hardware fail-safe — the protection that does not depend on anyone remembering to act, because it cannot be reasoned around.

The epistemic keeper: the layer that preserves the right to doubt — the person or process that can still ask whether the system’s output is true, and be heard when the answer is no.

Most of the failures in this book begin the same way. One of these keepers is removed, and the system that remains is trusted to hold the line alone.

This book is about what happens when the keeper is removed.

The Removal

Lighthouses were automated throughout the twentieth century. The last manned lighthouse in England was automated in 1998. The last in the United States was Boston Light, automated in 1998. The last keeper in Scotland left Flannan Isles in 1971. Country by country, rock by rock, the keepers departed and the machines took over.

The case for automation was overwhelming and, on its own terms, correct. Automated lights are more reliable than human-maintained lights. They do not require expensive support logistics — supply ships, fresh water, food, relief crews. They do

not expose human beings to the brutal conditions of offshore rocks. They do not create the psychological toll documented in generations of keepers: depression, alcoholism, the slow erosion of sanity in isolation. The cost per year of operation dropped by an order of magnitude. The failure rate dropped as well.

On every measurable metric, automation was superior.

Yet the automated systems could not see a ship in distress and radio for help. The keepers could and did — lighthouse keepers were responsible for hundreds of documented rescues over the centuries, pulling survivors from wrecks, launching lifeboats in conditions no sane person would enter, and providing shelter to shipwrecked sailors who reached the rock alive. The automated systems could not observe unusual weather patterns and warn approaching vessels by signal. They could not detect a malfunction in the light that was subtle enough to pass the mechanism's self-check but visible to a human eye — a lens slightly fouled, a rotation slightly irregular, a beam whose pattern had shifted just enough to confuse rather than guide.

The automation was superior on every measurable metric. The keeper was superior on the metrics that could not be measured — because the situations that required judgment were, by definition, the situations that were not anticipated, and you cannot measure the value of handling a situation you did not know would occur.

The Pattern

I am going to trace a pattern through this book that I believe is among the most important and least discussed dynamics in the history of technology.

The pattern is this:

A system is built. The system includes a human in the loop — a person whose job is to exercise judgment when the system encounters conditions it was not designed for. The system works. It works well. It works so well that the human's contribution becomes invisible, because the human is only needed in the rare cases when the system fails, and the system rarely fails.

Someone calculates the cost of the human. Someone compares it to the cost of the system operating without the human. The human is expensive. The system is cheap. The human is removed.

For a while — sometimes years, sometimes decades — nothing bad happens. The system operates without the human and performs as well as it did with the human, because the conditions remain within the range the system was designed for. The removal is declared a success. The savings are celebrated. The human is forgotten.

Then the system encounters a situation it was not designed for.

And there is no one there.

The consequence is proportional to the stakes.

In a lighthouse, a ship may go unwarned. In an aircraft cockpit, 228 people die over the Atlantic. In a radiation therapy room, a patient receives a hundred times the intended dose. In a criminal courtroom, an algorithm sentences a person based on the color of their neighborhood. In a military targeting system, a weapon selects a civilian structure because the pattern matched a probability threshold and no human was empowered to say: wait.

This is not a book about machines failing. All machines fail. Failure is an engineering problem, and engineering problems have engineering solutions. This is a book about the systematic removal of the human who could catch the failure — and the specific, documented, often fatal consequences that follow.

What I Will Show

Across twelve chapters, I will trace this dynamic through five domains: the systems that move us, the systems that heal us, the systems that judge us, the systems that watch us, and the systems that fight for us. In each domain, the same five patterns recur.

The Reliability Trap: the more reliable the automated system, the less practiced the human becomes at the task, and the worse the human performs when the system fails. The autopilot that flies perfectly for a thousand hours produces a pilot who cannot fly the thousand-and-first.

The Single Point of Trust: catastrophic failures trace, with remarkable consistency, to a system that trusted one sensor, one algorithm, or one assumption — without independent verification. One data source. One chance to be wrong. One failure away from disaster.

The Invisible Bias: when human decisions are encoded into automated systems, the biases travel with them — but stripped of the human context that might have recognized and corrected them. The bias becomes mathematics. Mathematics does not look biased. It looks objective. That is what makes it dangerous.

The Speed-Judgment Tradeoff: automated systems operate faster than human thought. As systems accelerate, the window for human intervention shrinks. At sufficient speed, the human is no longer in the loop. The human is beside it — a spectator to decisions that have already been made.

The Accountability Gap: when a human makes a catastrophic error, the chain of responsibility is clear. When an automated system does, responsibility diffuses — across designers, operators, manufacturers, regulators, and the system itself. Often, nobody is held accountable. The error is attributed to the system. The system is not a person.

The consequence falls on the victim. The lesson is not learned.

Trust is the mechanism of removal, but trust is not one thing. Across these cases it takes five forms, and telling them apart matters, because each fails differently.

Reliability trust: the system has worked before, so it is expected to work again — the trust of the operator who has crossed the same water a hundred times without incident.

Institutional confidence: an organization assumes its own processes, products, or oversight are sufficient — that someone, somewhere, has already checked.

Operational dependence: the human can no longer perform the task without the system, because the system has done it so consistently that the skill has atrophied.

Algorithmic deference: a decision-maker treats an output as authoritative because it looks quantified, neutral, and objective — and disagreeing with a number feels like disagreeing with arithmetic.

Trust transfer: an institution leans on a system not because it is better but because deferring to it lowers cost, friction, and visible blame. The system becomes the place responsibility goes to disappear.

Why This Book Exists

I wrote this book because the pattern would not leave me alone. The same architecture of failure — the keeper removed, the system trusted, the catastrophe arriving at the margin — repeated in every domain I examined: aviation, medicine, criminal justice, technology, warfare.

The question that unifies every chapter is the one that makes this pattern matter to every individual: when the machine is wrong — and it will be wrong, not once but repeatedly, not in some distant future but in the systems you depend on right now — will there be a human present, empowered, and skilled enough to catch the failure before it costs a life?

This book is about people. Specific people in specific moments: a pilot over the Atlantic at two in the morning with an autopilot that has just disconnected. A patient on a table beneath a radiation beam that the software has miscalibrated. A defendant in

a courtroom facing a sentence that an algorithm calculated from data the defendant cannot see or challenge. A family in a building that an autonomous targeting system has identified as a military objective. The people in these stories did not consent to the removal of the keeper.

Most did not know the keeper had been removed. They trusted the system because the system had been reliable, because the institution behind the system was credible, and because nobody told them that the human who used to stand between them and the machine's errors was no longer there.

Thomas Howell kept the light burning because he understood that the ships could not see whether he was suffering.

The ships could only see the light.

This book is for the ships — for the people who depend on systems they cannot see inside, maintained by keepers they do not know have been removed.

“This book is for the ships — for the people who depend on systems they cannot see inside, maintained by keepers they do not know have been removed.”

The Question

The automated lighthouse at Smalls still operates. The light is powered by solar panels and batteries. The lens rotates on electric motors. The system is monitored remotely from the mainland. It has been decades since a keeper stood on the rock.

The light is more reliable now than Thomas Howell could have maintained it.

The point is that there will be nights when the mechanism fails. Not a night. Nights.

There will be storms the designers did not model, corrosion the sensors did not detect, conditions that fall outside the parameters the system was built to handle. The failures will not announce themselves. They will arrive quietly, in the gap between what the system was designed for and what actually happens, and they will keep arriving for as long as the system operates.

Because the gap is permanent. No system, however sophisticated, can anticipate every condition it will face. That is not a limitation of engineering. It is a property of the world.

The question this book asks is not whether the machines will fail. They will, and they will fail more than once, and the failures will come at the moments when the stakes are highest, because those are the moments that test the boundaries of what the

system was designed for.

The question is whether, when those nights come — and they will keep coming, for as long as we build machines and trust them with consequential tasks — there will be a keeper watching the light.

CHAPTER TLK-2

The Unsinkable Ship

On the Titanic, the iceberg warnings, and the cost of confidence in engineering

The Unsinkable Ship

On the Titanic, the belief that sufficient engineering eliminates the need for caution, and the oldest form of the reliability trap.

Everyone knows the Titanic struck an iceberg. Almost everyone misunderstands why. The standard telling is an engineering story: the ship was poorly designed, the steel was brittle, the rivets were weak, the watertight compartments were not watertight enough. These details are largely accurate and entirely beside the point. The Titanic did not sink because the engineering failed. The engineering performed exactly as it was designed to perform. The hull withstood the impact along its length. The watertight doors closed automatically. The pumps activated.

The systems worked. The ship still sank, because the humans who operated it had made a series of decisions, hours and days before the collision, that the best engineering in the world could not compensate for. The Titanic sank because the people responsible for it trusted the machine more than they trusted the warnings.

The Warnings

On April 14, 1912 — the day of the collision — the Titanic received at least six ice warnings from other ships. The messages came through the ship's Marconi wireless room, a small cabin staffed by two radio operators: Jack Phillips and Harold Bride. The operators were employed not by the White Star Line but by the Marconi Company, and their primary responsibility was handling passenger telegrams — wealthy passengers sending and receiving personal messages from the middle of the Atlantic. The ice warnings were incidental to their main workload.

Some of the warnings reached the bridge. Some did not. The one from the SS Mesaba, received at approximately 9:40 PM — less than two hours before the collision — described a large field of ice directly in the Titanic's path. It was not delivered to the bridge because Phillips was busy clearing a backlog of passenger telegrams. He acknowledged the message and placed it under a paperweight on his desk. It was found there after the sinking.

Here is the first keeper failure, and it has nothing to do with engineering. The information existed. The warning was received. It was specific, timely, and accurate.

But the system — the organizational system, not the ship — did not treat ice warnings as a priority. They were one data stream among many, handled by operators whose primary incentive was passenger satisfaction, not navigation safety.

The keeper was present.

The keeper received the warning.

The keeper put it under a paperweight because the system he operated in did not empower him to act on it.

The Speed

The Titanic was traveling at approximately 22.5 knots — nearly its maximum speed — through waters known to contain ice. This was not an oversight. Captain Edward Smith was aware of the ice warnings that had reached the bridge. He adjusted the ship's course slightly southward. He did not reduce speed. The decision not to reduce speed was consistent with standard maritime practice of the era and with the institutional culture of the White Star Line. Slowing down meant arriving late. Arriving late meant disappointing passengers and the press.

The Titanic's maiden voyage was a commercial event as much as a nautical one, and the incentive structure — the pressure from the company, the expectations of the passengers, the competition with Cunard Line for the North Atlantic record — favored speed over caution. Captain Smith had thirty-eight years of experience at sea. He was the most senior captain in the White Star fleet. He had navigated ice before, many times, without incident. His experience told him that ice could be spotted in time to steer around it, that the lookouts would do their job, that the ship's size and construction provided a margin of safety that smaller vessels did not enjoy.

His experience was extensive, and it was wrong. The mechanism recurs throughout this book, and it deserves a precise name. Recklessness is the wrong word for it. Captain Smith was not reckless. He was experienced, competent, and operating within

the norms of his profession.

What he was, specifically, was *calibrated to a system that had never failed*. He had crossed the Atlantic with ice warnings before. He had maintained speed before. Nothing bad had happened before. His judgment was formed by a track record of success, and that track record of success produced a confidence that the next crossing would be like the last.

The keeper was present, experienced, and empowered — and his judgment was calibrated to a machine that had never failed him, so he trusted the machine more than he trusted the warning.

“Captain Smith was not reckless. He was calibrated to a system that had never failed. His judgment was formed by a track record of success. And a track record of success produces a confidence that the next time will be like the last.”

The Boats

The most telling detail about the Titanic disaster is the lifeboats, not the collision. The Titanic carried twenty lifeboats with a total capacity of 1,178 people. There were 2,224 people aboard. The ship could carry roughly half its passengers and crew to safety. The other half had no provision. This was not an accident and it was not illegal. The British Board of Trade regulations governing lifeboat capacity had been written in 1894, when the largest ships were under 10,000 gross tons. The Titanic was 46,328 gross tons.

The regulations had not been updated. Under the rules as they stood, the Titanic actually exceeded the minimum lifeboat requirement. It was, by the standards of its regulatory framework, overprovided with lifeboats.

But the regulatory failure is only part of the story. The deeper truth is that the White Star Line’s original plans for the Titanic included forty-eight lifeboats — enough for everyone aboard. Alexander Carlisle, the managing director of the shipyard that built the Titanic, designed the davit system to accommodate this number. The decision to carry twenty instead of forty-eight was made by the White Star Line’s managing director, J. Bruce Ismay, who judged that the additional boats would clutter the deck and alarm passengers.

The reasoning was explicit: a ship this well-engineered did not need that many lifeboats. The lifeboats were a concession to regulation, not a reflection of genuine risk. They were carried because the law required them, not because anyone expected

to use them. Here is the reliability trap in its purest pre-digital form: the engineering was so good that the safety measure was perceived as unnecessary. The backup was removed — not physically, but functionally, by reducing it below the level that would actually serve its purpose — because the primary system was trusted to make the backup irrelevant.

The Night

At 11:40 PM on April 14, 1912, lookout Frederick Fleet spotted the iceberg from the crow's nest and struck the warning bell three times. He telephoned the bridge: "Iceberg, right ahead." First Officer William Murdoch ordered the helm hard to starboard and the engines reversed. The ship began to turn. It was not enough. The Titanic struck the iceberg along its starboard side below the waterline, opening the hull across approximately 250 feet. Water flooded five of the sixteen watertight compartments. The ship was designed to remain afloat with four compartments flooded.

Five was one too many. Thomas Andrews, the ship's designer, who was aboard for the maiden voyage, calculated that the ship would sink in approximately one to two hours. He was correct. The lookouts, it should be noted, were operating without binoculars. The ship's binoculars had been locked in a cabinet, and the officer who had the key had been reassigned before departure and had taken the key with him.

Nobody had replaced the binoculars or forced the cabinet open. On a calm, moonless night — the worst conditions for spotting ice, because there are no waves to break against the berg and no moonlight to illuminate it — the lookouts were scanning the horizon with their naked eyes.

The keeper was in the crow's nest.

The keeper was doing his job.

The keeper was denied the tool that would have given him the seconds of additional warning that might have made the difference — because the tool was locked in a cabinet and nobody had thought it mattered enough to fix.

The Pattern, Named

The standard Titanic narrative is a morality tale about hubris — arrogant men who built an unsinkable ship and were punished by nature for their arrogance. That narrative is satisfying and incomplete. The Titanic was not brought down by arrogance in the theatrical sense. It was brought down by a perfectly rational, perfectly

understandable, and perfectly lethal form of trust. The engineering was trusted, so the lifeboats were reduced. The ship's track record was trusted, so the speed was maintained. The lookout system was trusted, so the missing binoculars were not replaced.

The wireless system was trusted, so the ice warnings were treated as low priority. At every decision point, the people involved were not ignoring the risk. They were evaluating the risk against the reliability of the system, and concluding — reasonably, based on experience — that the system would handle it.

The system handled it right up until the night it didn't.

And 1,517 people died because the preparations for that night had been made by people who did not believe it would come. This book is about the systematic removal of the human who could catch the failure, and the Titanic is its foundation stone, because it demonstrates that the pattern does not require a computer. It does not require artificial intelligence. It does not require any technology more sophisticated than a steam engine and a wireless telegraph.

It requires only a system that works well enough, for long enough, that the humans who depend on it stop preparing for the night it doesn't.

“The Titanic was not brought down by arrogance. It was brought down by a perfectly rational, perfectly understandable, and perfectly lethal form of trust. The people involved were not ignoring the risk. They were evaluating it against a system that had never failed.”

The Bridge

The Titanic went down in 1912.

The first electronic computer was built in 1944. Between these two dates, the pattern I have described — the reliability trap, the single point of trust, the keeper present but not empowered — repeated itself through industrial disasters, aviation accidents, medical errors, and military miscalculations. The details varied. The structure did not.

What changed with computers — and what makes the next ten chapters different from this one — is scale and speed. The Titanic's officers had hours to process ice warnings. A modern autopilot disconnects in milliseconds. The Titanic's lifeboat capacity was a decision made in a boardroom over weeks. An algorithm sentences a defendant in seconds.

The human failures aboard the Titanic were failures of judgment, operating at human speed, with human time to reconsider. The failures in the chapters that follow are the same failures of judgment, encoded into machines that operate at a speed that leaves no room for reconsideration. Computers did not invent the reliability trap. They automated it.

And an automated trap springs faster than a human can react.

CHAPTER TLK-3

The Invisible Interlock

On the Therac-25 radiation therapy machine and the software safety interlock that was removed

The Invisible Interlock

On the Therac-25 radiation therapy machine, the software that replaced a physical safety mechanism, and the patients who died because a timing error could only be triggered by an operator who was good at her job.

A radiation therapy machine is, in principle, a straightforward device. It generates a beam of high-energy particles — electrons or X-rays — and directs that beam at a tumor inside a patient's body. The beam destroys the tumor cells. The skill is in the precision: the beam must be strong enough to kill the tumor and narrow enough to spare the healthy tissue surrounding it. Too little radiation and the tumor survives. Too much and you destroy the patient along with the disease. The difference between a therapeutic dose and a lethal dose is not a matter of kind. It is a matter of degree.

The same machine that saves a life at 200 rads will end one at 20,000. The margin between healing and killing is a number on a screen, set by an operator, controlled by software, and delivered by a machine that the patient cannot see, cannot understand, and has no choice but to trust. Between June 1985 and January 1987, a radiation therapy machine called the Therac-25 delivered massive overdoses to at least six patients at clinics in the United States and Canada.

The overdoses ranged from approximately 13,000 to 25,000 rads — roughly one hundred times the intended dose.

Three patients died from the radiation injuries. The others suffered severe and permanent harm: radiation burns that penetrated deep into tissue, neurological damage, and the slow, agonizing progression of radiation sickness in organs that had received doses designed for industrial sterilization, not human therapy.

The machine's operator console displayed the message "MALFUNCTION 54" after the overdoses. The operators did not know what Malfunction 54 meant. The manual did not explain it. The manufacturer's technical support could not immediately identify it. In at least one case, an operator, seeing the malfunction message but believing the machine had simply failed to deliver the dose, reactivated the beam and irradiated the patient a second time.

"The difference between a therapeutic dose and a lethal dose is not a matter of kind. It is a matter of degree. The margin between healing and killing is a number on a screen."

The Predecessor

To understand what went wrong with the Therac-25, you must first understand what went right with the Therac-20. The Therac-20 was the previous model in the same product line, manufactured by Atomic Energy of Canada Limited (AECL). It was a dual-mode machine: it could deliver either a low-energy electron beam directly to the patient or a high-energy X-ray beam produced by directing the electrons at a tungsten target. The two modes required different configurations of the machine's internal components — in electron mode, a scanning magnet spread the beam evenly; in X-ray mode, a metal target and a flattening filter shaped the beam for deeper penetration.

The critical point is this: in X-ray mode, the electron beam was enormously powerful — powerful enough to penetrate the tungsten target and produce X-rays.

If that full-power beam were delivered directly to a patient without the target in place, the dose would be lethal. The Therac-20 prevented this with hardware interlocks — physical mechanisms that made it mechanically impossible for the machine to fire the full-power beam unless the target and flattening filter were correctly positioned. These interlocks operated independently of the software. They were physical barriers. They could not be overridden by a software error, because they were not controlled by software. They were controlled by gravity, springs, and mechanical linkages that either allowed or blocked the beam based on the physical position of the components.

The Therac-20 had software too. The software controlled the treatment parameters, the operator interface, and the sequencing of the machine's operations.

But the software did not control the safety-critical function — ensuring that the right components were in place before the beam fired. That function was performed by the hardware, independently, every time, regardless of what the software did. A

door with two locks: a deadbolt and an electronic lock. The deadbolt works by physical mechanism — a metal bar slides into a metal frame. The electronic lock works by software — a computer checks a code and signals a motor.

If the electronic lock malfunctions, the deadbolt still holds the door shut. The two systems are independent. The failure of one does not compromise the other. The Therac-20 was a door with two locks.

The Upgrade

The Therac-25 was the next-generation machine. It was faster, more versatile, and more computer-controlled than the Therac-20. It used a newer, more powerful processor. It offered a more sophisticated operator interface.

And it removed the hardware interlocks. The designers reasoned that the software could perform the safety checks that the hardware interlocks had previously provided. The software already controlled the treatment parameters. It already knew which mode the machine was in. It could verify, through software logic, that the correct components were in place before firing the beam. The hardware interlocks were, in this reasoning, redundant — an expensive, mechanical duplication of a function that the software already performed. The door was given one lock instead of two.

The deadbolt was removed because the electronic lock was believed to be sufficient. This decision was not made carelessly. It was made by engineers who understood the system they were building. The software had been derived, in significant part, from the Therac-20's software, which had operated for years without a safety-critical failure. The reasoning was the Titanic's reasoning, sixty-five years on: the system has worked.

The system has worked for a long time. The backup is expensive and, given the system's track record, unnecessary. Remove the backup.

The Bug

The software had a flaw. It was a type of error called a race condition — a situation where the outcome depends on the precise timing of two operations that happen almost simultaneously. It is worth explaining in plain terms, because understanding it is essential to understanding why the patients died. The Therac-25's operator interface allowed the technician to type in the treatment parameters: the mode (electron or X-ray), the energy level, and the dose.

After entering the parameters, the operator could review them on screen and, if a mistake was noticed, move the cursor back to the incorrect field and change it. This was a normal, expected workflow — the kind of correction any operator makes dozens of times a day. The race condition occurred when the operator changed the mode from X-ray to electron (or vice versa) very quickly — within approximately eight seconds of the initial entry.

If the mode was changed fast enough, the software updated the display to show the new mode but did not complete the internal reconfiguration of the machine's physical components. The screen told the operator the machine was in electron mode.

The machine was still configured for X-ray mode — with the full-power beam active and no target in place to absorb it. The beam fired. The patient received the full force of the X-ray-mode electron beam, directly, without the tungsten target, without the flattening filter, at a dose roughly one hundred times higher than intended.

The Cruel Irony

The detail that makes this story structurally important: The race condition could only be triggered by a fast operator. An operator who typed slowly — who took more than eight seconds to notice a mistake and correct it — would never encounter the bug. The software would have time to complete the internal reconfiguration before the correction was entered.

The machine would work correctly. An experienced operator — someone who had used the machine many times, who was skilled and efficient, who could spot a mistake and correct it in seconds rather than fumbling through the interface — was the only operator who could trigger the lethal error. The bug punished competence. It was invisible to the novice and lethal to the expert.

One of the patients, a man named Ray Cox, was treated at a clinic in Tyler, Texas, in March 1986. The operator entered X-ray mode, then quickly corrected to electron mode.

The machine displayed “MALFUNCTION 54.” Cox reported feeling a jolt — “like an electric shock,” he said — and pain in his shoulder. The operator, believing the machine had simply malfunctioned without delivering the dose, reactivated the beam. Cox received a second massive overdose. He died months later from the radiation injuries. The operator had no way to know what had happened. The screen said the dose had not been delivered. The manual did not explain what Malfunction 54 meant. The manufacturer had not disclosed that any hazard of this kind was possible.

The operator did the reasonable thing — the thing the system’s design led her to do. She trusted the screen.

The Investigation

When the accidents were eventually investigated — by the FDA, by independent researchers, and most thoroughly by Nancy Leveson and Clark Turner in what became one of the most cited papers in the history of software engineering — the findings revealed a pattern of failures that extended far beyond the race condition itself. AECL, the manufacturer, had tested the software.

But the tests were designed to verify that the software performed its intended functions correctly — not that it failed safely when something unexpected happened. The race condition was never tested because it arose from an interaction between operator behavior and software timing that the testers did not anticipate. You cannot test for a failure you cannot imagine. The software had been reused from the Therac-20 — but the context had changed.

On the Therac-20, the same software flaw may well have existed.

But it didn’t matter, because the hardware interlocks would have prevented the beam from firing regardless. The software bug was dormant on the Therac-20 — present in the code but harmless, because the hardware keeper was standing between the bug and the patient.

When the hardware keeper was removed in the Therac-25, the dormant bug became lethal. The mechanism now has a name: a latent defect that is harmless as long as an independent safety layer exists, and lethal the moment that layer is removed. The defect does not announce itself when the safety layer is taken away. It waits. It waits for the specific combination of conditions that activates it — and those conditions may not arrive for months or years.

The system appears to work perfectly during the interval. The removal of the safety layer is declared a success.

And then a patient sits on a table and an operator who is good at her job types a correction in under eight seconds.

The Institutional Failure

After the first reported overdose, at Kennestone Regional Oncology Center in Georgia in June 1985, the hospital’s physicist investigated and concluded that the Therac-25 had delivered an overdose. He contacted AECL. The manufacturer investigated and

could not reproduce the error. AECL concluded that the machine was safe and resumed operations.

After the second overdose, at the Ontario Cancer Foundation in Hamilton, Ontario, in July 1985, the pattern repeated. The hospital reported the incident. AECL investigated. AECL could not reproduce the error. AECL concluded the machine was safe.

After the third overdose, at the same Hamilton clinic in December 1985, AECL issued a software fix that addressed one potential cause. The fix did not address the race condition, which had not yet been identified. The fourth, fifth, and sixth overdoses occurred in 1986 and 1987, after the “fix” had been applied. Patients continued to be irradiated at lethal doses. The manufacturer continued to express confidence in the machine’s safety. It was not until the FDA intervened, conducted its own investigation, and ordered a series of corrective actions that the full scope of the defect was identified and addressed.

Six patients. Over a period of approximately two years. The information that the machine was dangerous existed after the first incident. The institutional system — the reporting channels, the manufacturer’s investigation process, the regulatory response — did not act on that information with the urgency the situation required, because the system trusted its own engineering.

The machine had been tested. The software had been verified. The Therac-20 had operated safely for years. The evidence that something was wrong was treated as anomalous rather than diagnostic — because the alternative, that the machine was fundamentally unsafe, was incompatible with the institution’s confidence in its own product. Captain Smith maintained speed through ice because the ship had never failed. AECL maintained confidence in the Therac-25 because the Therac-20 had never failed.

The pattern is the same.

The system’s track record of success produced an institutional resistance to believing that the system could fail catastrophically.

And while the institution resisted, patients continued to be harmed.

“Six patients. Over two years. The information that the machine was dangerous existed after the first incident. The institutional system did not act, because trusting its own engineering was easier than accepting that the engineering was killing people.”

The Lesson That Was Not Learned

The Therac-25 accidents led to significant reforms in medical device software regulation. The FDA strengthened its requirements for software verification. AECL redesigned the machine with independent hardware safety interlocks — restoring the keeper that had been removed. The Leveson-Turner paper became required reading in software engineering courses. For a time, the lesson seemed to have penetrated: do not remove independent safety layers. Do not trust software to do what hardware used to guarantee. Do not assume that a system is safe because its predecessor was safe in a different configuration.

And then the lesson was not learned.

In 2018 and 2019, a Boeing 737 MAX — a system that trusted a single sensor to control an automated flight function, with no independent hardware override, in a configuration that the pilots were not fully briefed on — killed 346 people in two crashes. The Therac-25's lesson, published in 1993, taught the world that removing independent safety layers and trusting software to replace them was a recipe for catastrophe. Twenty-five years later, the same architecture killed more than a hundred times as many people. That detail separates this book from a catalog of historical accidents.

The Therac-25 story did not end in the 1980s. The same pattern repeats — in aviation, in autonomous vehicles, in AI-driven medical diagnostics, in military targeting systems — because the economic incentive to remove the independent safety layer (it is expensive, it is slow, it is redundant most of the time) is always stronger than the institutional memory of what happened the last time it was removed.

The keeper is expensive.

The keeper is slow.

The keeper is unnecessary 99.99% of the time.

But the keeper is the difference between a system that fails and recovers and a system that fails and kills.

And the Therac-25 proved that in the starkest possible terms, in a treatment room where a patient trusted a machine to heal them and the machine delivered a hundred times the dose because an operator was good at her job and a deadbolt had been replaced by a line of code.

The Pattern Restated

The Titanic trusted its engineering, so the lifeboats were reduced. The Therac-25 trusted its software, so the hardware interlocks were removed. In both cases, the independent safety layer was eliminated because the primary system was judged to be sufficient. In both cases, the judgment was formed by a track record of success under conditions that did not include the specific failure that eventually occurred. In both cases, people died in the gap between what the system was designed for and what actually happened. The difference between the Titanic and the Therac-25 is the role of the computer.

On the Titanic, every failure was a human decision: maintain speed, reduce lifeboats, ignore warnings.

On the Therac-25, the lethal failure was a software timing error that no human could perceive, diagnose, or prevent in the moment it occurred. The patient on the table had no warning. The operator at the console had no indication. The error happened inside the machine, at the speed of the processor, in the eight-second window between one keystroke and the next. The Therac-25 removed a hardware interlock. The next machine in this story removed something more fundamental: the pilot's proficiency at the task the machine had taken over. And when the machine failed and the human was asked to take over, the skill was no longer there.

PART II: THE AUTOMATION PARADOX

On what happens when the machine gets good enough to be trusted — and the human forgets how to check

The Keeper Who Saved the World

Shortly after midnight on September 26, 1983, Lieutenant Colonel Stanislav Petrov was the duty officer at Serpukhov-15 — the Soviet Union's early-warning command centre, a bunker south of Moscow responsible for monitoring American nuclear missile launches. His job was simple in description and annihilating in consequence: if the system detected an incoming strike, he was to report it immediately to his superiors, who would have minutes to decide whether to launch a retaliatory strike before Soviet weapons were destroyed on the ground. At 12:14 AM, the system reported a launch.

A single American Minuteman intercontinental ballistic missile, inbound. The screen flashed LAUNCH in large red letters. The siren sounded.

Within seconds, the system reported a second missile.

Then a third.

Then a fourth and fifth. Five American nuclear missiles, confirmed by satellite, heading toward the Soviet Union. Petrov had the training, the authority, and the obligation to report the attack as confirmed.

The system was telling him, with high confidence, that the United States had launched a first strike. His commanding officers were waiting. The protocol was clear.

The machine had spoken. He did not report it. He declared it a false alarm. His reasoning was human in a way that no algorithm could replicate. A genuine American first strike, he judged, would not begin with five missiles. It would begin with hundreds — an overwhelming salvo designed to destroy Soviet retaliatory capability before it could be used. Five missiles made no strategic sense.

And the ground-based radar, which should have corroborated the satellite detection, showed nothing.

The system was confident. Petrov was not. He trusted his judgment over the machine's certainty, knowing that if he was wrong, the missiles would arrive in approximately twenty minutes and his country would have no warning. He was right. The satellite system had misidentified sunlight reflecting off high-altitude clouds as missile exhaust plumes. It was a sensor error — the kind of failure that machines produce and humans catch, when the humans are allowed to exercise judgment rather than merely ratify a computation.

“The system was confident. Petrov was not. He trusted his judgment over the machine’s certainty, knowing that if he was wrong, the missiles would arrive in twenty minutes and his country would have no warning.”

Stanislav Petrov saved the world because he was a keeper who was allowed to keep. He had time — minutes, not seconds. He had context — training in how a real attack would look versus what the machine was showing him. He had authority — the power to override the system based on his own assessment.

And he had something no machine possesses: the capacity to recognise that the data, however confidently presented, did not make sense. The chapters that follow are about what happens when those conditions are removed.

When the time is compressed from minutes to seconds.

When the context is eroded by automation that deskills the human.

When the authority is overridden by a system the operator cannot see inside.

When the machine's confidence becomes, structurally, unchallengeable.

In 1983, Petrov had twenty minutes and his own judgment. In the Lavender targeting system deployed forty years later, the human in the loop has twenty seconds and a rubber stamp. The distance between those two numbers is the distance this section measures.

* * *

CHAPTER TLK-4

The Handoff

On the automation paradox in aviation and the shrinking window for human judgment

The Handoff

On Air France Flight 447, fewer than four minutes over the Atlantic, and the most dangerous moment in any automated system.

At 2:10 AM local time on June 1, 2009, the autopilot on Air France Flight 447 disconnected. The Airbus A330 was at 35,000 feet over the equatorial Atlantic, roughly three and a half hours into a flight from Rio de Janeiro to Paris.

Two hundred and twenty-eight people were aboard — 216 passengers and 12 crew. The captain, Marc Dubois, had left the cockpit to rest. In the left seat was first officer David Robert, 37 years old. In the right seat was first officer Pierre-Cédric Bonin, 32 years old. It was dark. The aircraft was passing through the Intertropical Convergence Zone — a belt of thunderstorms that circles the equator — and had entered a region of high-altitude ice crystals. The ice crystals blocked the aircraft's pitot tubes — small probes on the outside of the fuselage that measure airspeed by detecting the pressure of oncoming air.

With the pitot tubes obstructed, the flight computers received inconsistent airspeed data. The autopilot, designed to disengage when its data is unreliable, did what it was designed to do. It handed the aircraft back to the pilots. This is the handoff. The moment when the automated system, having encountered a condition outside its operating parameters, transfers control to the human. The moment the training wheels come off. The moment the keeper must do the thing the machine has been doing for them, in conditions the keeper has never practiced, with no warning and no preparation.

Three minutes and twenty-eight seconds later, the aircraft hit the surface of the Atlantic Ocean at approximately 11,000 feet per minute. All 228 people aboard were

killed.

“The autopilot disconnected. Three minutes and twenty-eight seconds later, 228 people were dead. Between those two events was a human being asked to do something the machine had made him forget how to do.”

The Three Minutes

The cockpit voice recorder and flight data recorder were recovered from the ocean floor two years after the crash, at a depth of nearly 13,000 feet. The French Bureau of Investigation and Analysis — the BEA — published a detailed reconstruction of the final minutes.

What the recorders revealed is a study in human capability collapsing under precisely the conditions the automation was supposed to prevent.

When the autopilot disconnected, Bonin — the pilot flying — was startled. His first instinct was to pull back on the side stick, pitching the nose of the aircraft upward. This is a natural human reaction to a perceived emergency — pull up, gain altitude, create distance from the ground. It is also, in the situation the aircraft was in, exactly wrong. An aircraft flies because air flowing over its wings generates lift.

If the nose is pitched too high, the angle of the wing relative to the oncoming air — called the angle of attack — becomes too steep. The smooth airflow over the wing breaks apart into turbulence. The wing stops generating lift. This is called an aerodynamic stall, and it is one of the most basic concepts in flight training. The recovery is equally basic: push the nose down to reduce the angle of attack, allow the wing to regain smooth airflow, and the aircraft flies again.

Every student pilot learns this in their first weeks of training. Push the nose down. Regain airspeed. Fly. Bonin pulled the nose up. The aircraft began to stall. The stall warning sounded — a loud, synthetic voice repeating the word “STALL” in the cockpit. Over the next three minutes, the stall warning sounded seventy-five times. Bonin continued to hold the nose up.

The Paradox

How could a trained airline pilot not recognize a stall? How could a pilot ignore seventy-five stall warnings? How could a pilot maintain an input that was killing the aircraft for three full minutes while the cockpit told him what was happening? The answer is the automation paradox, and it is the central concept of this chapter. A

modern Airbus autopilot is extraordinarily reliable. It manages the aircraft's speed, altitude, heading, and engine power with a precision that no human pilot can match over sustained periods.

On a typical long-haul flight, the autopilot may be engaged for 95% or more of the flight time. The pilots monitor instruments, manage communications, and handle the takeoff and landing. The flying — the actual physical manipulation of the aircraft through the air — is done by the computer. This arrangement is, on average, safer than having humans fly the entire time. Autopilots do not get tired, do not get distracted, and do not make the small control errors that accumulate during hours of manual flying.

The statistical case for automation in aviation is overwhelming. Commercial aviation is the safest form of transportation in human history, and automation is a primary reason.

But here is what the statistics conceal. Every hour the autopilot flies is an hour the pilot does not fly. Every hour the pilot does not fly is an hour of proficiency that erodes. The skills required to hand-fly an aircraft — to sense the aircraft's state through the controls, to recognize subtle deviations, to respond instinctively to unusual attitudes — are *perishable* skills. They degrade without practice in the same way that a musician's technique degrades without rehearsal. The hands forget. The instincts dull.

The connection between what the pilot feels and what the aircraft is doing frays. The automation paradox states: the more reliable the automated system, the less practiced the human operator becomes at the task, and the worse the human performs when the system fails and they must take over. The paradox is that the very reliability of the automation — the thing that makes it valuable — is the mechanism that degrades the human capability that is needed when the automation is gone. Bonin had approximately 2,900 flight hours.

He had been trained in stall recovery. He had passed his checkrides. On paper, he was qualified.

But the vast majority of his flying hours had been spent monitoring an autopilot, not hand-flying an aircraft at the limits of its flight envelope, at night, over the ocean, with unreliable instruments. The training was in his file. The proficiency was not in his hands.

The Confusion

The cockpit voice recorder reveals a scene of escalating confusion. Robert, in the left seat, did not initially realize that Bonin was holding the nose up.

On the Airbus, the two side sticks operate independently — unlike traditional yokes, which are mechanically linked so that one pilot can see and feel what the other is doing.

When Bonin pulled back, Robert's stick did not move. Robert had no tactile indication that the other pilot was commanding a nose-up input. This design choice matters. In an aircraft with mechanically linked controls, one pilot's input is immediately visible and physically felt by the other pilot. The connection is a form of communication — a continuous, wordless awareness of what the other person is doing. The Airbus side sticks removed this connection in favor of a design that was lighter, more compact, and compatible with the fly-by-wire system that the autopilot required.

The tradeoff was that the two pilots could give contradictory inputs simultaneously without either one knowing. For three minutes, that is exactly what happened. Bonin pulled the nose up. Robert, who eventually realized the aircraft was descending, pushed the nose down. The computer algebraically summed their inputs — the opposing commands partially canceling each other out. The aircraft descended. Captain Dubois, awakened by the initial upset, returned to the cockpit approximately one minute and thirty seconds after the autopilot disconnected. He found his two first officers struggling with contradictory instruments, contradictory inputs, and a stall warning that kept sounding and stopping — stopping because at very low speeds, the aircraft's computers determined that the airspeed data was unreliable and disabled the stall warning, creating the perverse situation in which the warning sounded when the aircraft was partially stalled and went silent when the stall was complete.

In one of the final exchanges captured by the voice recorder, Bonin said: "But I've had the stick back the whole time." It was the first moment the other pilots understood what he had been doing. Robert replied: "No, no, no, don't climb." Four seconds later, the aircraft struck the water.

The Keeper Who Could Not Keep

There were three keepers in the cockpit of Air France 447.

Three trained, licensed airline pilots.

The keeper was not absent.

The keeper was not distracted.

The keeper was not negligent.

The keeper was *deskilled* — a word that the aviation safety community uses to describe the degradation of manual flying ability that results from prolonged reliance on automation. The Therac-25 removed a hardware interlock — a physical safety mechanism. Air France 447 revealed the removal of something less visible and more consequential: the pilot's proficiency at the task the machine had taken over. The interlock was a device. The proficiency was a human capability.

And the capability was not removed in a single decision, the way the interlock was removed in the Therac-25 redesign. It was eroded gradually, flight by flight, hour by hour, as the autopilot performed the task and the pilot watched. This is what makes the automation paradox more insidious than a simple design flaw. The Therac-25's missing interlock was a decision that could be traced to specific engineers at a specific company. The degradation of Bonin's flying skills was not a decision anyone made. It was the cumulative consequence of a system that worked as designed.

Nobody decided to deskill the pilots. The automation deskilled them by doing its job too well.

The Reliability Trap Restated

Air France 447 is the most vivid expression of a pattern that aviation safety researchers have documented for decades.

In 2013, the FAA published a study that found airline pilots were increasingly unable to fly effectively when the autopilot was unavailable — not because of inadequate training, but because the opportunity to practice had been systematically removed by the automation itself. Researchers in human factors engineering have a name for this dynamic: the reliability trap. A system becomes more reliable.

Because it is more reliable, the human operator practices less.

Because the human practices less, the human's ability degrades.

Because the ability degrades, the consequences of the system failing become more severe. The very success of the automation is the mechanism of the human's decline. The trap is that you cannot escape it by making the system *more* reliable — that only tightens the trap.

And you cannot escape it by making the system *less* reliable — that defeats the purpose of the automation. The reliability trap reaches every domain where automation performs a task that a human must take over when the automation fails: power plant operations, surgical robotics, autonomous vehicle systems, naval warfare

systems. Every domain where the machine is better than the human 99% of the time, and the human must be better than the machine in the 1% that matters most.

The asymmetry is structural.

The machine practices constantly.

The human practices never.

And the test comes without warning.

The Question the Industry Does Not Ask

The BEA's final report on Air France 447 was published in 2012. It identified multiple contributing factors: the frozen pitot tubes, the crew's failure to recognize and respond to the stall, the design of the independent side sticks, and the stall warning logic that went silent at full stall. The report led to changes in pilot training, including mandatory high-altitude stall recovery exercises and increased emphasis on manual flying proficiency. These were important and necessary reforms. They were also insufficient, because they addressed the symptoms without confronting the structural question: if the autopilot is reliable enough that pilots rarely fly manually, and manual flying skills degrade without practice, and the autopilot is designed to disconnect at precisely the moments when conditions are most difficult and manual skills are most needed — then the system contains a contradiction that no amount of training can resolve. The automation is reliable 99.99% of the time. The handoff occurs during the 0.01% when conditions are worst. The pilot must perform a task they rarely practice, in conditions they have never experienced, with no transition and no warning.

The system is designed so that the human is needed most precisely when the human is least prepared.

“The system is designed so that the human is needed most precisely when the human is least prepared. The automation paradox is not a bug. It is a structural property of every system that automates a skill.”

The Pattern Deepens

The Titanic's officers had hours to process warnings and chose to trust the ship. The Therac-25's operators had seconds to respond to an error message they could not interpret. Air France 447's pilots had fewer than four minutes to recover an aircraft from a condition that the automation had, in a meaningful sense, trained them not to

recognize. There is a counterexample that makes the mechanism precise. On July 20, 1969, during the lunar descent of Apollo 11, the guidance computer became overwhelmed with data and threw a series of memory alarms — codes 1202 and 1201 — while steering the lunar module toward a boulder-strewn crater.

Neil Armstrong took manual control, flew the craft laterally over the hazard, and landed with roughly fifteen seconds of fuel remaining. He could do this because he had not been deskilled by the system he was overriding. He was a test pilot who had practiced manual control thousands of times.

When the machine failed, the keeper's hands remembered what to do. The pilots of Air France 447 had not practiced manual flight at altitude in years.

When their machine disconnected, their hands did not remember. The difference between Armstrong and Bonin is not courage. It is practice.

And practice is the first thing that automation erodes.

The time available for human judgment is shrinking. The demands on that judgment are increasing.

And the skill required to exercise it is degrading, because the automation that was supposed to assist the human is instead replacing the human's ability to perform without assistance. The Boeing 737 MAX did not merely fail to help the pilot. It actively fought the pilot.

The keeper was in the cockpit, hands on the controls, pulling with everything he had.

The machine overrode him.

And the machine was acting on data from a single sensor that was wrong.

CHAPTER TLK-5

The Override

On the Boeing 737 MAX, the MCAS system, and the pilots who could not override the machine

The Override

On the Boeing 737 MAX, a single faulty sensor, and 346 people who died because the machine decided it knew better than the pilot.

In 2011, Boeing had a problem. Its rival, Airbus, had launched the A320neo — an updated version of the bestselling A320 narrow-body jet, fitted with new, more fuel-efficient engines. The A320neo offered airlines approximately 15% better fuel economy, which over the twenty-to-thirty-year life of a commercial aircraft represents hundreds of millions of dollars. Airlines were placing orders. American Airlines, a longtime Boeing customer, signaled it was considering switching to Airbus. Boeing faced a choice. It could design an entirely new aircraft — a clean-sheet design with modern aerodynamics and systems — which would take approximately a decade and cost tens of billions of dollars.

Or it could do what Airbus had done: take an existing, proven airframe and fit it with new engines. The 737, Boeing's equivalent to the A320, had been in production since 1967. It was the bestselling commercial aircraft in history. Boeing knew it. Airlines knew it. Pilots knew it. Regulators knew it. Fitting new engines to the 737 would be faster, cheaper, and — crucially — would allow Boeing to certify the new aircraft as a variant of the existing 737 rather than as a new type.

This meant that airlines could operate the new aircraft without expensive pilot retraining. A pilot certified on the 737 could step into the MAX and fly it the same day. Boeing chose the second option. The aircraft that resulted was the 737 MAX.

The Physics Problem

The new engines were larger and heavier than the ones they replaced. The 737's airframe sits relatively low to the ground — a design feature that dates to the 1960s, when the aircraft was designed to operate from poorly equipped airports where passengers boarded via stairs rather than jet bridges. The low ground clearance meant the larger engines could not be mounted in the same position as the old ones. They would scrape the runway. Boeing's engineers moved the engines forward and upward on the wing.

This solved the ground clearance problem but introduced an aerodynamic one: in certain flight conditions, particularly at high angles of attack and low speeds, the repositioned engines created additional lift ahead of the aircraft's center of gravity. This had the effect of pitching the nose upward — the opposite of what the pilot intended when pulling back on the yoke.

If uncorrected, this nose-up tendency could lead to a stall. The aerodynamic behavior was subtle. In most flight conditions, the 737 MAX handled identically to the earlier 737 NG. The nose-up tendency appeared only in a narrow band of the flight envelope — high angle of attack, low speed, specific flap configurations.

But it was there, and it was a problem, because if the MAX handled differently from the NG in any condition, it might require a separate type rating. That would mean pilot retraining. That would mean airlines would need to invest in new training programs, new simulators, new certification processes. That would eliminate the MAX's key competitive advantage: operational interchangeability with the existing 737 fleet.

“If the MAX handled differently from the NG in any condition, it would need a separate type rating. That would eliminate the aircraft's key competitive advantage. The solution was software.”

The Software Solution

Boeing's solution was a software system called the Maneuvering Characteristics Augmentation System — MCAS.

The system was designed to monitor the aircraft's angle of attack and, when it detected the nose-up tendency caused by the repositioned engines, automatically push the nose down by adjusting the horizontal stabilizer — the large movable surface at the tail of the aircraft that controls pitch. In principle, MCAS was a reasonable engineering solution. Software augmentation of flight characteristics is common in modern aircraft. The Airbus fly-by-wire system uses software extensively to shape the

aircraft's handling characteristics. The concept was not new.

But the implementation had three features that, in combination, made MCAS lethal. First, MCAS relied on a single angle-of-attack sensor. The 737 MAX has two angle-of-attack vanes — one on each side of the fuselage. MCAS used only one of them.

If that sensor was faulty, MCAS would receive incorrect data and act on it. There was no cross-check, no comparison between the two sensors, no redundancy. A single point of failure controlled a system that could override the pilot's inputs. Second, MCAS could activate repeatedly. In its original design specification, MCAS was intended to provide a single, limited nose-down correction. During development, the system's authority was increased — the amount of stabilizer movement it could command grew larger, and MCAS was allowed to activate repeatedly, each time adding more nose-down input.

If the faulty sensor continued to report a high angle of attack, MCAS would continue to push the nose down, each cycle adding to the cumulative deflection, until the stabilizer reached its mechanical limit. The cumulative force was strong enough to overpower the pilots' ability to pull back on the yoke. Third, the pilots were not told about MCAS. Boeing did not include MCAS in the 737 MAX's pilot training materials.

The system was not described in the aircraft's flight manual. Pilots transitioning from the 737 NG to the 737 MAX were not briefed on MCAS, were not trained to recognize its activation, and were not taught how to disable it.

The system existed, it could override their control of the aircraft, and they did not know it was there.

Lion Air Flight 610

On October 29, 2018, Lion Air Flight 610 departed Jakarta, Indonesia, bound for Pangkal Pinang. The aircraft was a 737 MAX 8, delivered to Lion Air just two months earlier. There were 189 people aboard. Twelve minutes after takeoff, the aircraft crashed into the Java Sea. There were no survivors. The flight data recorder revealed the following sequence. Shortly after takeoff, the left angle-of-attack sensor began reporting erroneous data — indicating a high angle of attack when the aircraft was flying normally. MCAS, reading this data, activated and pushed the nose down.

The captain, feeling the unexpected nose-down movement, pulled back on the yoke to correct it. He succeeded temporarily.

Then MCAS activated again. The captain pulled back again. MCAS pushed down again. This cycle repeated more than twenty times in the eleven minutes between the first MCAS activation and the crash. Each time, MCAS added more nose-down stabilizer. Each time, the captain fought to bring the nose back up. The force required to hold the nose up increased with each cycle as the stabilizer accumulated more deflection. The captain was in a tug-of-war with a machine that had more authority than he did, that could not be reasoned with, that could not be told the sensor was wrong, and that he did not know the name of.

The first officer searched the quick reference handbook for the source of the problem. He could not find it, because MCAS was not in the handbook. The aircraft hit the water at approximately 450 knots in a near-vertical dive. The impact was so severe that the remains of the passengers could only be identified through DNA analysis.

Ethiopian Airlines Flight 302

On March 10, 2019 — less than five months later — Ethiopian Airlines Flight 302 departed Addis Ababa bound for Nairobi. The aircraft was a 737 MAX 8. There were 157 people aboard, from 35 countries. Six minutes after takeoff, the aircraft crashed into terrain. There were no survivors. The flight data recorder revealed the same pattern. A faulty angle-of-attack sensor. MCAS activation. The captain fighting the nose-down commands. MCAS overriding him. The stabilizer accumulating deflection beyond what the pilots could counteract. Captain Yared Getachew was 29 years old with over 8,000 flight hours.

Unlike the Lion Air crew, Getachew and his first officer had the benefit of the worldwide attention that followed the Lion Air crash. Boeing had issued a bulletin after Lion Air reminding pilots of the existing runaway stabilizer procedure. Getachew and his first officer followed that procedure: they flipped the cutout switches that disconnected the electric stabilizer trim, which should have disabled MCAS.

But with the electric trim disabled, the pilots had to move the stabilizer manually using a hand crank in the center pedestal. The problem was that MCAS had already driven the stabilizer to a significant nose-down position. At the aircraft's speed, the aerodynamic forces on the stabilizer were so great that the manual crank could not overcome them. The pilots could not physically turn the wheel. They were strong enough. The physics were not.

In a last, desperate attempt, the crew re-engaged the electric trim, hoping to use its motor to drive the stabilizer back. MCAS immediately activated again and pushed the nose further down. The aircraft entered an unrecoverable dive.

The keepers did everything right. They identified the problem. They followed the published procedure. They flipped the correct switches.

And they died because the machine had already moved the control surface beyond the point where human strength could recover it, and nobody at Boeing had tested whether the published procedure was physically executable after MCAS had been active for more than a few seconds.

The Architecture of the Failure

The US House Transportation Committee’s investigation, published in 2020 after an 18-month inquiry, documented a failure that extended from the engineering floor to the boardroom to the regulatory agency that was supposed to provide independent oversight. Boeing had designed MCAS to solve a business problem as much as an engineering one. The aerodynamic issue was real, but the decision to solve it with a software patch rather than a redesign was driven by schedule pressure and the need to maintain the 737 MAX’s certification status as a variant of the 737 NG.

Every design choice — single sensor, no pilot notification, expanded authority — was made in the direction of minimizing the regulatory and training impact of the system. The FAA, the regulatory body responsible for certifying the aircraft as safe to fly, had delegated significant portions of the certification process to Boeing itself. This was not unusual — the FAA had used delegated certification for years as a way of managing its workload.

But the effect was that Boeing employees were simultaneously designing the system and approving the design. The engineers who determined that MCAS was safe were, in many cases, the same engineers who had designed it. The independent safety layer — an external regulator with the authority and expertise to challenge Boeing’s assessment — had been thinned to the point of transparency. Boeing’s own internal communications, revealed during the Congressional investigation, showed that some employees recognized the risks.

One test pilot wrote in an internal message that MCAS was “running rampant” in a simulator session. Another employee, discussing the decision not to include MCAS in pilot training, expressed concern about the system’s potential behavior. These concerns did not change the certification outcome. The schedule held. The aircraft was

delivered. It was the Therac-25's institutional reflex, scaled upward. AECL could not reproduce the error and declared the machine safe. Boeing's analysis concluded MCAS was safe. In both cases, the manufacturer's confidence in its own engineering overrode the evidence — and in both cases, an independent safety layer that should have caught the error had been weakened: hardware interlocks removed in the Therac-25, regulatory independence compromised in the 737 MAX.

The pattern does not vary. The institution trusts itself. The backup is eroded. People die in the gap.

Removal, Degradation, Override

The three chapters that form Part II of this book trace an arc that defines the relationship between humans and automated systems. In the treatment room, the keeper was removed. The Therac-25's hardware interlocks — independent, physical, reliable — were taken away because the software was judged to be sufficient.

The human operator was present but had no means of detecting or preventing the error. The safety layer between the bug and the patient was gone. In the cockpit, the keeper was degraded. Air France 447's pilots were at the controls, trained and licensed.

But the automation that was supposed to assist them had, over thousands of hours, eroded the manual flying skills they needed when the automation departed.

The keeper was present but had been rendered unable to keep. In this chapter, the keeper was overridden. The 737 MAX pilots were present, aware that something was wrong, and actively fighting to save the aircraft.

The machine fought back.

The machine had more authority than the humans.

The machine was acting on a single faulty data point.

And the machine won. Removal. Degradation. Override. Each step is more insidious than the last, because each step leaves the human with the *appearance* of control while reducing the *substance* of control. The Therac-25 operator appeared to control the treatment but could not detect the error. The Air France pilots appeared to be flying but had forgotten how. The 737 MAX pilots were fighting for their lives and losing to a system they did not know existed.

“Removal. Degradation. Override. Each step leaves the human with the appearance of control while reducing the substance of control. The human is first absent, then unprepared, then powerless.”

Beyond the Cockpit

The next three chapters leave the world of physical systems — machines, aircraft, medical devices — and enter the world of algorithmic systems. The failures in Part II killed people through physics: radiation beams, aerodynamic stalls, nose-down forces that exceeded human strength. The failures in Part III will harm people through decisions: who goes to prison, who receives medical care, who is shown content that radicalizes them toward violence. The mechanism changes.

The pattern does not. In every case, an automated system replaces or overrides human judgment. In every case, the system operates on data that is incomplete, biased, or wrong.

The humans who are harmed have no way to see inside the system, challenge its reasoning, or appeal its decision.

And the keeper — the human who should exercise independent judgment at the point of consequence — has been removed, degraded, or overridden. The cockpit is a constrained environment. The angle of attack is a physical measurement. The stall is a physical phenomenon. The crash is a physical event. The next chapters enter a domain where the inputs are social, the outputs are human lives, and the errors are not crashes but injustices — quieter, slower, distributed across millions of individual decisions, and for exactly those reasons, harder to see and harder to stop.

CHAPTER TLK-6

The Orbit

On the Ariane 5 rocket, reused software, and the assumptions that survive their context

The Orbit

On the systems that exceeded our capacity to verify them.

The previous chapters described systems where the keeper was removed and the failure was, in retrospect, legible. The Titanic’s officers maintained speed through ice — a decision traceable to a culture of trust in engineering. The Therac-25’s race condition was a specific flaw in specific code, found and fixed once investigators knew where to look. Air France 447’s pilots could not fly the plane — a skill gap produced by years of reliable automation. The 737 MAX fought its pilots because of a single-sensor dependency that Boeing chose not to disclose.

In each case, the failure had a name. A cause. A line of code, a design decision, a regulatory gap that could be pointed to and said: *there*.

The keeper was removed, and the removal was identifiable. The lesson, however painfully learned, was learnable. This chapter is about what happens when the failure stops being legible.

When the systems we build grow past the point where human beings can verify that they will work correctly under all the conditions they will encounter.

When the keeper may be present, empowered, skilled, and willing — but the system is too complex for the keeper to understand. This is a different kind of threat than a removed keeper. It is a keeper who *cannot* keep, because the thing being kept has crossed a threshold beyond which human comprehension does not reach.

Thirty-Seven Seconds

On June 4, 1996, the European Space Agency launched the Ariane 5 rocket on its maiden flight from the Guiana Space Centre in Kourou, French Guiana. The rocket

carried four Cluster satellites — a scientific mission to study the interaction between the solar wind and the Earth’s magnetosphere. The total cost of the payload and launch vehicle was approximately \$500 million. Thirty-seven seconds after liftoff, the rocket veered off course, broke apart under aerodynamic stress, and self-destructed. The payload was lost. The Cluster mission had been ten years in planning.

Hundreds of scientists across fourteen European countries had designed the instruments, calibrated the sensors, and waited for the data that would advance humanity’s understanding of space weather. The principal investigators had spent careers building toward this launch. They watched from the control room in Toulouse as a decade of work disintegrated in thirty-seven seconds. The mission was eventually reflown — Cluster II launched successfully in 2000 — but the four years of delay, the loss of irreplaceable instruments, and the human cost of watching your life’s work explode are not captured in the technical reports.

The subsequent investigation, led by an inquiry board chaired by Jacques-Louis Lions, identified the cause with unusual precision. The failure began in the rocket’s inertial reference system — the component that measures the rocket’s orientation and velocity. The software in the inertial reference system attempted to convert a 64-bit floating-point number, representing the rocket’s horizontal velocity, into a 16-bit signed integer. The number was too large to fit. The conversion produced an overflow error. The error handler, following its programmed instructions, shut down the inertial reference system.

The backup inertial reference system — the redundant unit, the safety layer — had already failed, 72 milliseconds earlier, from the identical error. Both units ran the same software. Both encountered the same overflow. The redundancy was an illusion: two copies of the same code, executing the same flaw, failing in the same way, 72 milliseconds apart.

“The backup system failed 72 milliseconds before the primary system, from the identical error. Two copies of the same code. The same flaw. The redundancy was an illusion.”

With both inertial reference systems offline, the flight computer received no valid orientation data. It interpreted the error codes from the failed systems as flight data — data indicating that the rocket needed a massive and immediate course correction. The nozzles of the solid rocket boosters swiveled to their maximum deflection. The rocket turned sharply. The aerodynamic forces exceeded the vehicle’s structural limits. The rocket broke apart. The self-destruct system activated.

The critical detail: the software that caused the failure was not written for the Ariane 5. It was reused from the Ariane 4, the previous rocket in the program. The Ariane 4 software had worked correctly for years. The conversion that failed on the Ariane 5 had never failed on the Ariane 4, because the Ariane 4's trajectory produced horizontal velocity values that were small enough to fit in a 16-bit integer. The Ariane 5's trajectory was different — steeper, faster, producing larger velocity values that exceeded the integer's range.

The code worked exactly as written. It was written for a different rocket.

The Review That Did Not Catch It

The Ariane 5 software was not unreviewed. It was tested. It was verified. The European Space Agency employed rigorous development and review processes. The reuse of Ariane 4 software was a deliberate decision, made on the reasonable basis that the software had a proven track record. The Lions report identified that the specific conversion from 64-bit floating-point to 16-bit integer had been analyzed during the Ariane 4 program and determined to be safe — because the Ariane 4's flight parameters guaranteed that the value would never exceed the 16-bit range.

When the software was transferred to the Ariane 5, this analysis was not repeated. The assumption that the value would remain within range was inherited along with the code. The assumption was valid for the old rocket. It was not valid for the new one. The reader who has been following this book will recognize the architecture. The Therac-25 reused software from the Therac-20. The software had a flaw that was harmless on the old machine because the hardware interlocks prevented it from manifesting.

On the new machine, the interlocks were removed, and the flaw became lethal. The Ariane 5 reused software from the Ariane 4. The software had a limitation that was harmless on the old rocket because the flight profile kept the values within range.

On the new rocket, the flight profile was different, and the limitation became catastrophic. In both cases, the system was reviewed. In both cases, the reviews did not catch the flaw. Not because the reviewers were incompetent, but because verifying that inherited code is safe in a new context requires understanding every assumption embedded in the inherited code and testing every assumption against the new context. In simple systems, this is feasible. In complex systems, the number of embedded assumptions exceeds the capacity of any review team to enumerate, let alone test.

The Unit

Three years after the Ariane 5 disaster, NASA lost the Mars Climate Orbiter — a \$328 million spacecraft designed to study the Martian atmosphere and serve as a communications relay for future missions. On September 23, 1999, the spacecraft approached Mars for its orbital insertion maneuver and was never heard from again. It had entered the atmosphere at an altitude of approximately 57 kilometers instead of the planned 226 kilometers — too low to survive. The investigation revealed that one engineering team, at Lockheed Martin, had been providing thruster performance data in imperial units — pound-force seconds.

The navigation team at NASA's Jet Propulsion Laboratory expected the data in metric units — newton-seconds. The difference is a factor of approximately 4.45. Over the nine months of the spacecraft's cruise to Mars, the accumulated error from this mismatch shifted the trajectory enough to destroy the mission. A unit conversion error. Not a novel algorithm. Not an unprecedented engineering challenge. A conversion between pounds and newtons — the kind of calculation a physics student learns in their first semester.

Three hundred and twenty-eight million dollars lost because one team said pounds and another team heard newtons, and no process in between caught the discrepancy. The Mars Climate Orbiter failure is, in isolation, almost comical — the kind of error that seems impossible at the level of sophistication involved.

But it illustrates a principle that the Ariane 5 failure also illustrates, and that this chapter exists to name: the failures in complex systems are rarely complex. They are simple errors — a data type conversion, a unit mismatch, an inherited assumption — embedded in systems so complex that the simple error is invisible.

The Complexity Threshold

The Ariane 5's flight software contained approximately several hundred thousand lines of code. This was considered complex at the time. It was reviewable — with difficulty, with significant effort, and with the gaps that the Lions report identified — but reviewable. A team of engineers could, in principle, read every line, trace every data flow, and test every conversion. The fact that they did not catch the fatal conversion was a failure of process, not of possibility. Modern software systems operate at a different scale.

The Linux kernel — the operating system that runs the majority of the world's servers, including those hosting cloud services, financial systems, and critical

infrastructure — contains over 30 million lines of code. A modern automobile contains approximately 100 million lines. The Google codebase is estimated at over 2 billion lines. No individual can read, understand, or verify these systems in their entirety. No team can.

The systems are maintained by organizations that understand their respective components, with no single person or group holding a complete understanding of the whole. These are systems whose behavior is, in the most literal sense, beyond human comprehension in totality. We can understand the parts. We can test the interfaces between parts. We can employ automated testing tools that exercise millions of code paths.

But the space of possible interactions between components in a system of 100 million lines of code is combinatorially vast — larger than the number of atoms in the universe. Testing covers a fraction. Review covers a fraction. The rest is trust.

The Black Box

And then there are neural networks. A large language model — the kind of AI system that powers modern chatbots, code generators, and an increasing number of decision-support tools in medicine, law, finance, and government — contains billions of parameters. These parameters are numerical values that were not written by engineers. They were learned during training — adjusted through millions of iterations of a mathematical optimization process that modifies the values to improve the system's performance on its training data.

The resulting system works. It produces outputs that are, much of the time, remarkably capable.

But no human designed those parameters. No human can explain, for a given output, exactly which parameters contributed to the result and why.

The system is not opaque because its creators chose to hide the logic. It is opaque because there is no logic in the human sense — no sequence of if-then rules, no decision tree, no flowchart. There are billions of numbers, arranged in layers, that collectively produce behavior. The behavior works. The mechanism is beyond human interpretation. The Ariane 5's fatal code was readable. An engineer could open the file, find the conversion, and see the flaw. The flaw was not found in time, but it was findable.

A neural network's equivalent of the Ariane 5 flaw — the embedded assumption that will produce catastrophic failure under conditions the system was not trained on

— is not readable. It is distributed across billions of parameters in ways that no current technique can fully trace.

The keeper who wants to verify the system cannot verify it. Not because the keeper lacks access, or skill, or time.

Because the system’s complexity has crossed a threshold beyond which human verification is not difficult but definitionally impossible.

The Implication

This chapter exists to establish a fact that the remaining chapters of this book must confront.

The systems now being deployed in the most consequential domains of human life — medical diagnosis, criminal sentencing, military targeting, financial markets, critical infrastructure — are systems that no human being can fully verify. Better tools will not resolve this condition, because the condition is structural — a property of the systems themselves. Calling a neural network with 175 billion parameters “difficult to interpret” understates the problem. Its behavior comes from a mechanism that has no human-readable explanation.

We can probe it. We can test it. We can observe its outputs and characterize its behavior statistically.

But we cannot look inside it the way Jacques-Louis Lions looked inside the Ariane 5’s code and said: here is the conversion, here is the overflow, here is the fault. The Therac-25’s flaw was a race condition. It was found, identified, and fixed. The 737 MAX’s flaw was a single-sensor dependency. It was found, identified, and fixed. The Ariane 5’s flaw was a data type conversion. It was found, identified, and fixed. In each case, the flaw was legible — expressible in language, traceable to a specific line of code or a specific design decision.

The fix was possible because the flaw was comprehensible.

What do we do when the flaw is not comprehensible?

When the system produces a wrong output — a misdiagnosis, a wrongful recommendation, a catastrophic financial trade — and no one can explain why?

When the keeper looks inside the system and finds not a readable error but an impenetrable matrix of numbers that produced the wrong answer for reasons that cannot be expressed in language? The chapters that follow will document what happens when systems of this complexity are deployed in domains where the stakes are measured in lives, liberty, and justice. Algorithms that sentence defendants.

Healthcare systems that allocate care. Recommendation engines that steer what billions of people believe. Targeting systems that select who dies. Surveillance infrastructure that monitors entire populations.

In each case, the keeper will face the same question this chapter has raised: how do you keep a system you cannot understand?

“How do you keep a system you cannot understand? The Ariane 5’s flaw was findable. A neural network’s equivalent flaw is distributed across billions of parameters. The system has crossed a threshold beyond which human verification is not difficult but impossible.”

What Has Been Established

Take stock of what this book has documented so far. A ship whose officers trusted the engineering more than the warnings, and 1,517 people died in the Atlantic. A radiation machine whose hardware safety layer was removed because the software was believed to be sufficient, and patients received a hundred times the intended dose. An aircraft whose pilots could not fly because the autopilot had made flying unnecessary, and 228 people fell from the sky. A second aircraft whose automated system fought its pilots to the death because it trusted a single faulty sensor.

And now: a rocket and a spacecraft destroyed not by malice or negligence but by the simple, inevitable consequence of building systems too complex for their creators to fully verify.

Every failure so far left a keeper who could, in principle, have understood the system — the captain who read ice warnings, the operator who saw Malfunction 54, the pilots who held the controls. The Ariane 5 and the Mars orbiter end that era. They are failures of complexity itself: the keeper cannot verify what the keeper cannot comprehend.

And the systems that follow in the remaining chapters of this book — the algorithms that judge, the engines that amplify, the machines that target, the infrastructure that watches — are systems built on exactly this kind of unverifiable complexity.

The keeper’s challenge is about to get harder. The light at Smalls still shines, more reliable than Thomas Howell could have kept it. But the mechanism was never the question. The night the mechanism fails is the question — whether anyone is watching, and whether the watcher can understand what has gone wrong.

CHAPTER TLK-7

The Sentencing Machine

On COMPAS, algorithmic sentencing, and the question of what objectivity means when the training data is biased

The Sentencing Machine

On the COMPAS algorithm, the courtrooms that trusted software to predict who would commit a crime, and the defendant who could not cross-examine the code that determined his future.

The previous three chapters told stories about machines that killed people. The beam fired. The aircraft crashed. The stabilizer drove the nose into the ground. In each case, the moment of failure was physical, immediate, and unmistakable. The patient felt the jolt. The cockpit voice recorder captured the final seconds. The wreckage was recovered from the ocean floor. This chapter tells a different kind of story.

Nobody dies.

Nobody is physically injured.

The machine does not malfunction.

The machine works exactly as designed, produces the output it was built to produce, and that output is used by human institutions to make decisions that alter the course of human lives — who goes to prison, for how long, under what conditions. The harm is not a beam or a crash. The harm is a number on a screen, presented to a judge, treated as evidence, and used to justify keeping a human being in a cage. The shift from Part II to Part III is a shift in what failure means.

In Part II, the machine failed and people died. In Part III, the machine succeeds — does exactly what it was designed to do — and people are harmed anyway, because the design itself encodes a harm that the designers did not intend and the operators cannot see.

The Promise

The idea behind algorithmic risk assessment in criminal justice is, on its surface, reasonable. Human judges are inconsistent. Studies have shown that sentencing outcomes vary with the judge's mood, the time of day, the order in which cases are heard, and — most troublingly — the defendant's race. A landmark 2018 study of federal sentencing data found that Black men received sentences roughly 19% longer than white men convicted of similar offenses, after controlling for criminal history and offense severity. Judicial discretion, applied inconsistently, produces injustice.

An algorithm, the argument goes, can do better. Feed it data — criminal history, demographic information, behavioral indicators — and it will produce a risk score: a number that predicts, with some statistical accuracy, the likelihood that a defendant will reoffend. The score is consistent. It does not depend on whether the judge had lunch. It does not depend on the defendant's appearance or demeanor. It produces the same output for the same input, every time. It is, in a word, objective. This was the promise of COMPAS — the Correctional Offender Management Profiling for Alternative Sanctions system — developed by a company called Northpointe (later Equivant).

COMPAS uses a 137-question assessment to generate risk scores for criminal defendants. The scores are presented to judges on a scale from 1 to 10, where higher numbers indicate higher predicted risk of recidivism. Judges in courtrooms across the United States use these scores to inform decisions about bail, sentencing, and parole. The promise was objectivity. The promise was consistency. The promise was that the machine would be fairer than the human.

And for a time, the promise was believed.

“The promise was that the machine would be fairer than the human. An algorithm cannot be racist. An algorithm does not have a mood. An algorithm does not skip lunch.”

The Investigation

In 2016, a team of journalists at ProPublica published an investigation that would become one of the most cited pieces of data journalism in history. Julia Angwin, Jeff Larson, Surya Mattu, and Lauren Kirchner obtained COMPAS risk scores for more than 7,000 defendants arrested in Broward County, Florida, between 2013 and 2014. They then tracked which of those defendants actually went on to commit new crimes

over the following two years, and compared the algorithm's predictions against reality. COMPAS was roughly as accurate as a coin flip for predicting violent recidivism — about 61% overall accuracy, which subsequent research showed was comparable to the prediction accuracy of untrained volunteers given nothing more than the defendant's age and number of prior convictions.

The algorithm's advantage over human judgment was, at best, marginal.

But the errors were not distributed equally. Black defendants who did not go on to reoffend were nearly twice as likely to be incorrectly classified as high risk compared to white defendants who did not reoffend. White defendants who did go on to reoffend were nearly twice as likely to be incorrectly classified as low risk compared to Black defendants who did reoffend. The algorithm was wrong in both directions — but it was wrong in a way that systematically overpredicted danger for Black defendants and systematically underpredicted danger for white defendants.

The algorithm did not use race as an input variable. Northpointe was emphatic on this point. Race was not in the questionnaire. Race was not a factor in the score. The algorithm was, in the company's framing, race-neutral. This is where the chapter arrives at the concept it exists to explain.

The Proxy

In statistics, a proxy variable is a variable that is not itself the thing you are measuring but that correlates closely enough with it to serve as a stand-in. Your zip code is not your race.

But in the United States, where decades of redlining, discriminatory lending, and housing segregation have sorted people into neighborhoods along racial lines, your zip code is a remarkably effective proxy for your race. Your employment history is not your race.

But in a country where the unemployment rate for Black Americans has been roughly double the rate for white Americans for every year since the Bureau of Labor Statistics began tracking it, your employment history carries racial information. Your number of prior arrests is not your race.

But in a country where Black Americans are arrested at 2.6 times the rate of white Americans — a disparity driven in significant part by differential policing of Black neighborhoods — your arrest record carries racial information. COMPAS did not use race. It used proxies for race.

And the proxies worked, in the sense that they produced scores that correlated with race in precisely the way that direct use of race would have. The algorithm achieved the outcome of racial discrimination without the mechanism of racial discrimination. The bias was invisible because it was structural, embedded in the correlations between variables rather than in any single variable. This is Pattern 3 — the Invisible Bias — in its purest form. The bias is not in the algorithm. The bias is in the data.

The data reflects the world as it is — a world built by centuries of racial discrimination in housing, employment, education, and policing. The algorithm learns the patterns in the data.

The patterns in the data include the effects of that discrimination. The algorithm reproduces the discrimination, not because it was instructed to, but because the discrimination is the pattern. Imagine teaching a child to sort photographs into two categories: “likely trustworthy” and “likely untrustworthy.” You never mention race.

But all the photographs labeled “untrustworthy” in your training set happen to be of people from the same neighborhood, which happens to be predominantly Black, which happens to be over-policed, which happens to produce more arrest records, which happens to be the training data the algorithm learned from. The child — and the algorithm — learns the correlation. Not because you taught the correlation.

Because the correlation was in the world you showed them.

The Defendant

In 2013, Eric Loomis was arrested in La Crosse, Wisconsin, for driving a car that had been used in a shooting. He was not accused of the shooting itself. He pleaded guilty to eluding a police officer and operating a vehicle without the owner’s consent. At sentencing, the judge cited Loomis’s COMPAS risk score — which rated him as high risk — as a factor in imposing a sentence of six years of initial confinement. Loomis challenged the sentence. His argument was straightforward: the COMPAS algorithm is proprietary.

Its methodology, its weighting of variables, and its validation data are trade secrets owned by Northpointe. Loomis could not see how his score was calculated. He could not challenge the algorithm’s assumptions. He could not present evidence that the algorithm was flawed. He could not, in the most fundamental sense, confront his accuser — because his accuser was a black box owned by a private company that refused to disclose how the box worked. The case reached the Wisconsin Supreme Court.

In 2016, the court ruled that the use of COMPAS in sentencing did not violate Loomis's due process rights. The court acknowledged that the proprietary nature of the algorithm was a concern. The court acknowledged that COMPAS scores correlate with race. The court acknowledged that the defendant could not fully understand or challenge the score.

And the court ruled that the score could be used anyway, as long as it was not the sole basis for the sentence. Read that again. The court said: we know the algorithm is opaque. We know it correlates with race. We know the defendant cannot challenge it. We will allow it anyway. The US Supreme Court declined to hear Loomis's appeal.

The Impossibility

In the aftermath of the ProPublica investigation, a remarkable academic debate unfolded — one that revealed something troubling not just about COMPAS but about the mathematical foundations of algorithmic fairness itself. Northpointe responded to ProPublica's findings by arguing that COMPAS was, in fact, fair. Their definition of fairness was different from ProPublica's. Northpointe argued that among defendants who received the same risk score, the actual recidivism rates were similar across racial groups. A Black defendant scored as high risk and a white defendant scored as high risk had approximately the same probability of actually reoffending.

The score meant the same thing regardless of race. This is called predictive parity. ProPublica's definition of fairness was different. They argued that among defendants who did not reoffend, Black defendants were far more likely to have been incorrectly classified as high risk. The error rate — the rate at which the algorithm was wrong — was not equal across racial groups. This is called equal false positive rates, or error rate balance.

Here is the devastating mathematical result: in 2016, researchers proved that in any population where the base rates of the predicted outcome differ between groups — which is to say, in any population where the groups are not identical — it is mathematically impossible to satisfy both definitions of fairness simultaneously. You cannot have predictive parity and equal error rates at the same time, unless the groups have identical base rates of the outcome being predicted. In the United States, Black Americans are rearrested at higher rates than white Americans.

This is a statistical fact. It is also a fact that is itself produced by the same system of differential policing, prosecution, and incarceration that the algorithm is supposed to transcend. The base rates are unequal because the system is unequal.

And because the base rates are unequal, it is mathematically impossible to build an algorithm that is fair by both reasonable definitions of fairness. This is not a software bug. This is not a fixable flaw. This is a mathematical proof that algorithmic fairness, in a society with unequal outcomes across groups, is a contradiction in terms. You must choose which kind of unfairness you prefer.

And the choice is moral, not technical. It is precisely the kind of decision that requires a keeper — a human being who understands the context, weighs the competing values, and takes responsibility for the outcome.

The Missing Keeper

In the courtroom, the keeper is the judge. A human judge is imperfect. A human judge is inconsistent. A human judge has bad days and blind spots and unconscious biases. All of this is true, and all of it was the justification for introducing algorithmic risk assessment in the first place.

But a human judge can be cross-examined. A human judge's reasoning can be reviewed on appeal. A human judge can be asked to explain why they believed this defendant was dangerous and that defendant was not. A human judge can look at a score and say: I know what the number says, but I have sat across from this person, I have heard their story, I have seen things the questionnaire does not capture, and I believe the number is wrong. A human judge can exercise judgment.

That is, in fact, why we call them judges. COMPAS does not eliminate the judge. The judge still pronounces the sentence.

But COMPAS shifts the epistemic ground.

When a judge has a number in front of them — a number produced by a system described as scientific, validated on data, endorsed by the corrections department — the burden shifts. The judge who follows the score needs no justification. The judge who overrides the score must explain why. The path of least resistance is to defer to the number.

And the number carries the biases of the world that produced it.

The keeper is still on the bench.

But the keeper's authority has been hollowed out.

The keeper has been given a tool that looks like evidence, feels like objectivity, and carries the weight of institutional endorsement — and overriding that tool requires the judge to assert their own fallible human judgment against the machine's output. The pilots of Air France 447 were present but deskilled. The judge in a COMPAS

courtroom is present but deauthorized — not formally stripped of power, but structurally discouraged from exercising it.

The Scale

COMPAS is no outlier — one system in an expanding apparatus of algorithmic decision-making in criminal justice. Risk assessment tools are used in pretrial detention decisions, sentencing, parole hearings, and probation conditions across the majority of US states. Some jurisdictions use tools developed by private companies. Others use tools developed by academic institutions or government agencies. The specific algorithms vary. The structural dynamics do not. The algorithm takes data that reflects the history of the criminal justice system — including its documented racial disparities — and uses that data to make predictions about the future.

The predictions are presented to human decision-makers as objective inputs.

The human decision-maker faces the same structural pressure: defer to the number, because overriding it requires justification and deference does not. The number of people affected is not small. In the United States alone, approximately 1.9 million people are held in prisons, jails, and detention facilities on any given day, and people enter jail more than seven million times each year. Algorithmic risk assessment touches a significant and growing fraction of these cases. Each case is a human being whose liberty is at stake. Each case is a decision that a machine has influenced and that the affected person cannot fully understand, challenge, or appeal.

The Deeper Problem

There is a question beneath the question of algorithmic fairness, and it is the question that matters. Should we predict crime at all? The entire premise of risk assessment — algorithmic or otherwise — is that we can identify, before the fact, who is likely to commit a crime and act on that prediction. The prediction is an assessment of probability, not a finding of guilt.

And acting on a probability — keeping someone in jail, denying bail, imposing a longer sentence — because of what they *might* do is fundamentally different from punishing them for what they *did*. The philosophical tradition that underlies the Western legal system is built on the principle that punishment follows action. You are punished for what you have done, not for what a statistical model predicts you will do. Risk assessment inverts this. It punishes the prediction.

And the prediction carries the inequalities it claims to transcend. This inversion matters, because the legal framework makes it unambiguous. The presumption of innocence is not a technicality. It is the bedrock of Western constitutional law — enshrined in Article 11 of the Universal Declaration of Human Rights, the foundation of the Fifth, Sixth, and Fourteenth Amendments to the United States Constitution. A person cannot be punished for something they have not done. COMPAS inverts this. A defendant who receives a high risk score serves additional time — longer pretrial detention, harsher sentencing, denied parole — based on a statistical prediction about their future behavior.

They are being punished not for a crime they committed but for a crime the algorithm predicts they *will* commit, based on characteristics they share with a demographic group. This is statistical profiling dressed as risk assessment. It is the presumption of guilt, computed.

Nobody is challenging this constitutionally, and the reason is the same reason that recurs throughout this book: the people affected lack the resources to litigate it. The populations most harmed by recidivism algorithms are disproportionately Black, disproportionately poor, and disproportionately represented by overworked public defenders who do not have the bandwidth to mount a constitutional challenge to the algorithmic infrastructure of the criminal justice system. The constitutional violation is hiding in plain sight because the victims cannot afford to make it visible.

And this logic does not stop at the courtroom. The same reasoning that says “this person is 60% likely to reoffend, so we keep them locked up longer” is the reasoning that says “this building is 70% likely to contain a combatant, so we strike it.” Both treat actual human beings as probabilities. Both invert the presumption of innocence.

One operates in a courtroom in Wisconsin. The other operates from a drone over Gaza. The constitutional violation is the same. The altitude is different. A human judge who admits that a defendant’s zip code influenced their sentencing decision would face immediate scrutiny. An algorithm that uses zip code as one of 137 variables produces a number that carries the same influence through a mechanism so complex that the influence is undetectable to the person who receives it. The judge sees a 7 out of 10.

The judge does not see the zip code inside the 7.

“The judge sees a 7 out of 10. The judge does not see the zip code inside the 7. The algorithm does not hide the bias. It dissolves the bias into a number, which is a more effective way of hiding it.”

The Pattern Crosses the Threshold

In the Therac-25, the operator could not see the race condition. In Air France 447, the pilots could not see each other's inputs. In the 737 MAX, the pilots could not see MCAS. The failure was invisible to the person who needed to prevent it. COMPAS applies this architecture to human liberty. The judge cannot see inside the algorithm. The defendant cannot see inside the algorithm. The public cannot see inside the algorithm. The algorithm's producer claims trade secret protection.

And the courts have ruled that this opacity is permissible.

But there is a difference, and it is the difference that makes Part III of this book more disturbing than Part II. The Therac-25 malfunctioned. The autopilot disconnected. MCAS received bad data. In each case, the system did something it was not supposed to do. COMPAS does exactly what it is supposed to do. It produces a risk score from the data it is given. The problem is not that the system fails. The problem is that the system succeeds — and the success encodes a harm that is built into the structure of the data, the structure of the question, and the structure of the society that produced both.

The keeper is not fighting a malfunction.

The keeper is fighting a system that is working as intended, backed by institutional authority, presented as objective, and resistant to challenge. That is a harder fight.

And it is the fight that defines the remaining chapters of this book.

But this chapter must be honest about a tension in its own argument. The COMPAS algorithm did not invent racial bias in sentencing. It inherited it.

The human judges whose decisions trained the algorithm were themselves biased — and their bias was invisible, unchallenged, and unquantified for decades. Keeping the human judge in the loop does not eliminate the bias. It preserves it in a form that is harder to measure and harder to correct. It was the algorithm — consistent, examinable, producing the same output from the same input — that made the bias *visible* for the first time. ProPublica could measure the racial disparity precisely *because* the algorithm produced measurable outputs.

The human judges' individual, opaque reasoning had never been subjected to equivalent scrutiny.

The case for keeping the human in the loop is therefore not a case for the status quo. It is a case for the human who can recognize the bias and correct it — which requires the bias to be exposed.

And sometimes it takes an algorithm to expose it.

The keeper's value is not neutrality.

The keeper is not neutral.

The keeper's value is the capacity for recognition and correction that no algorithm possesses on its own.

But there is a harder counterpoint, and intellectual honesty requires naming it. On April 26, 1986, at the Chernobyl Nuclear Power Plant in Ukraine, human operators running a safety test deliberately disabled the automated emergency cooling systems, overrode hardware interlocks, and pushed the reactor into an unstable state that caused the explosion. At Chernobyl, the keepers were not overridden by the machine.

The keepers overrode the machine — and the machine's constraints were the only thing that had been preventing catastrophe.

The keeper is not inherently safe. Human judgment is only valuable if the human is exercising wisdom.

When the keeper acts with arrogance, with recklessness, with the conviction that they know better than the system they are disabling, the automated constraint becomes the last line of defense.

This book argues for the keeper. Chernobyl reminds us that the argument is conditional.

From the Courtroom to the Clinic

COMPAS decides who is dangerous. The next chapter tells the story of an algorithm that decides who is sick — and the patients it systematically determined were healthier than they actually were, because the data it was trained on confused spending money on healthcare with needing healthcare.

In a country where Black patients have historically received less care, the algorithm learned that Black patients need less care. The algorithm was wrong. The patients suffered.

And the bias, once again, was invisible to the humans who trusted the output. The sentencing machine predicted futures. The diagnostic will misread the present.

CHAPTER TLK-8

The Diagnostic

On the healthcare algorithm that confused cost with need, and the racial bias embedded in the data

The Diagnostic

On a nephrologist in Scarborough who looked beneath the categories, the algorithm that confused cost with need, and the question of what we measure when we think we are measuring health.

Scarborough sits on the eastern edge of Toronto. Sixty percent of its residents are immigrants. Seventy-seven percent belong to an ethnic group other than Canada's White majority. It is one of the most diverse urban populations in the Western world, and it is home to one of North America's largest dialysis programs — annually serving over 6,000 patients with chronic kidney disease and more than 1,200 on dialysis. Dr. Tabo Sikaneta is a nephrologist at the Scarborough Health Network. He has spent two decades treating kidney disease in this population, and in 2025 he published a study in *BMJ Open* that should change how every healthcare system thinks about the data it feeds its algorithms.

The headline finding: within Canada's universal healthcare system, dialysis prevalence was 4.2 times higher among immigrants than non-immigrants.

But Dr. Sikaneta did not stop at the headline. He disaggregated. He looked beneath the category of "immigrant" and found that the disparity was not uniform. It was concentrated in patients born in specific regions — the Caribbean, Southeast Asia, and South Asia. Kidney function declined fastest in patients from the Caribbean, South Asia, and East Asia.

And the finding that forecloses the simplest explanations: *the year of immigration did not influence the outcomes*. A patient who arrived from the Caribbean thirty years ago showed the same disease trajectory as one who arrived five years ago. Acculturation does not explain it. Recent arrival does not explain it. And cost of

access — the insurance barrier that distorts healthcare data in the United States — cannot explain it, because Canada’s universal system removes that barrier. The disparity persists in a universal system, across decades of residency, concentrated in specific populations whose risk factors — genetic predisposition, epigenetic markers, environmental exposures in the country of origin, dietary and metabolic patterns that persist across generations — are not captured by any of the standard categories an algorithm would use.

An algorithm that sorted patients by “immigrant versus non-immigrant” would flag millions of healthy people as high risk while telling clinicians nothing about where to intervene. An algorithm that sorted by ethnicity would get closer but would still miss the birth-country specificity that Dr. Sikaneta’s disaggregation revealed. Only a clinician with deep knowledge of the population — who knows which communities carry elevated risk, and that the risk persists regardless of how long the patient has lived in Canada — can look at a patient’s chart and ask: is the category this system is using adequate to the complexity of this patient’s risk? That is the keeper function.

And this chapter is about what happens when nobody performs it — when the algorithm is trusted, the categories are accepted, and the wrongness of the metric is invisible to everyone in the system.

“The algorithm sorted patients by a category. The keeper looked beneath the category. The difference, for thousands of patients, was whether anyone asked if the category itself was adequate.”

The Wrong Metric

In the courtroom, an algorithm encoded the biases of the criminal justice system and used them to predict who would commit a crime. The bias entered through the training data — arrest records, conviction rates, socioeconomic variables — that reflected a system of policing built on racial inequality. The output looked objective. The input was not. This chapter tells a story that is, in some ways, more troubling. COMPAS was built by a private company selling a product to courtrooms. The algorithm at the center of this chapter was built by a health system trying to help its patients.

The intentions were good. The designers were careful.

The system was validated against its own metrics and performed well.

And it was systematically denying care to Black patients who needed it, not because anyone instructed it to, but because the metric it was optimizing was the wrong metric — and the wrongness was invisible to everyone until a team of

researchers decided to look.

The Algorithm

In 2019, Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan published a paper in *Science* that would reshape the conversation about algorithmic bias in healthcare. They examined a widely used commercial algorithm — deployed across hospitals and insurance systems to manage the care of approximately 200 million Americans — that was designed to identify patients who would benefit from enrollment in high-risk care management programs. These programs provide additional resources: dedicated nurse coordinators, more frequent check-ins, priority access to specialists. They are expensive and limited in capacity.

The algorithm decided who got in. The algorithm used a reasonable-sounding proxy for health need: healthcare costs. Patients who had spent more money on healthcare in the previous year were predicted to need more care in the following year. The logic was straightforward — if you spent a lot on healthcare last year, you are probably sick, and you will probably need a lot of healthcare next year. Cost as a proxy for need. The algorithm worked. On its own terms, it performed well. It identified patients with high costs and predicted their future costs with reasonable accuracy.

It was validated, deployed, and trusted by the institutions that used it. There was one problem. The proxy was wrong.

The Gap

In the United States, healthcare spending tracks far more than health need: insurance coverage, access to providers, geographic proximity to medical facilities, trust in the medical system, the ability to take time off work, transportation, language barriers, and — pervasively, measurably, and historically — race. Black Americans, on average, spend less on healthcare than white Americans with comparable health conditions. Black Americans are not healthier; they face greater barriers to accessing care: lower rates of insurance coverage, fewer nearby providers, more experiences of discrimination in medical settings that discourage return visits, and the cumulative legacy of a medical system that, within living memory, experimented on Black patients without consent, denied Black patients treatment, and maintained segregated hospitals. The reader who began this chapter in Scarborough will recognize the deeper problem. Dr. Sikaneta's research demonstrated that the disparity persists even when

the cost barrier is removed — within universal healthcare, across decades of residency, concentrated in populations whose risk factors are not captured by the categories any algorithm would use.

The Obermeyer algorithm's proxy was cost.

But the Canadian evidence shows that the problem runs deeper than the proxy. The data itself carries the disparity. Fix the billing and you still have patients whose needs the standard categories understate. The algorithm saw cost. The algorithm did not see the gap between cost and need. A Black patient who needed care but did not receive it — because they could not afford it, could not reach it, or did not trust the system enough to seek it — appeared in the data as a low-cost patient.

Low cost, to the algorithm, meant low need. Low need meant the patient did not qualify for the care management program that might have addressed the barriers keeping them from care in the first place.

The result, quantified by Obermeyer and his colleagues, was stark: at a given risk score, Black patients were considerably sicker than white patients with the same score. The researchers found that Black patients assigned the same level of predicted risk had 26% more chronic conditions than their white counterparts. The algorithm was not identifying the sickest patients. It was identifying the most expensive patients.

And in a country where access to healthcare is unequal, the most expensive patients and the sickest patients are not the same people. The researchers estimated that correcting the bias would increase the percentage of Black patients in the high-risk care management program from 17.7% to 46.5%. Nearly half of the Black patients who should have been receiving additional care were being denied it — not by a racist decision-maker, but by a metric that confused spending with suffering.

The Mirror

The COMPAS story and the healthcare story are structurally identical. In both cases, the algorithm used a proxy variable that correlated with the outcome of interest but also correlated with race, because the underlying system that produced the data was marked by racial inequality. In COMPAS, the proxy was prior arrests (correlated with both reoffending and race through differential policing). In healthcare, the proxy was cost (correlated with both health need and race through differential access to care).

But the healthcare case is sharper in one specific respect. COMPAS was built by a private company selling a risk assessment tool to courtrooms.

One can argue — and many have — that the commercial incentives were misaligned with justice. The healthcare algorithm was built by a health system trying to allocate scarce resources to the patients who needed them most. The intent was to help. The designers chose cost as a proxy because cost data is abundant, reliable, and easy to collect, while direct measures of health need — like the number and severity of chronic conditions — are harder to gather consistently at scale. The choice was pragmatic and defensible.

Cost data is clean. Health data is messy. The algorithm worked on the data it had. The problem was not carelessness or malice. The problem was that the data it had was a mirror of a society in which healthcare access is distributed unequally, and the mirror reflected the inequality as faithfully as it reflected everything else. That fidelity is what makes the Invisible Bias so difficult to fight. The bias is not in the algorithm. The bias is in the world. The algorithm is a mirror.

The mirror shows what is in front of it.

If what is in front of it is a society where Black patients receive less care, the mirror will show Black patients needing less care. The mirror does not lie.

But it does not tell the truth, either. It tells a version of the truth that depends on the angle at which it is held — and the angle, in this case, is determined by the metric the designers chose.

The Face

The same year the Obermeyer paper was published, a different study had already been making headlines for two years — one that demonstrated the same pattern in a domain where the stakes are different but the mechanism is identical.

In 2018, Joy Buolamwini, a researcher at the MIT Media Lab, and Timnit Gebru, then at Microsoft Research, published a paper called “Gender Shades.” They evaluated three commercial facial recognition systems — from Microsoft, IBM, and Face++ — on their ability to correctly classify faces by gender.

The results revealed a failure so systematic that it could not be attributed to random error. For lighter-skinned males, the systems achieved error rates as low as 0.8%. For darker-skinned females, error rates reached 34.7%. The gap was not small. It was a factor of more than forty.

The systems that technology companies were marketing as accurate and ready for deployment were, for a significant portion of the human population, barely functional. The reason was the training data. Facial recognition systems learn to identify faces by

being shown millions of photographs during training. The datasets that these systems were trained on were disproportionately composed of lighter-skinned faces, particularly lighter-skinned male faces.

The systems became very good at recognizing the faces they saw most often and very bad at recognizing the faces they rarely saw. The bias was not in the algorithm's logic. It was in the algorithm's education — in the examples it was given to learn from.

The Deployment

A facial recognition system that fails 0.8% of the time for one group and 34.7% of the time for another is not merely inaccurate.

When deployed in law enforcement — which it has been, across hundreds of police departments in the United States and governments around the world — it becomes a machine for generating false identifications that fall disproportionately on the people the system sees least clearly.

In 2020, Robert Williams, a Black man in Detroit, was arrested at his home in front of his wife and daughters for a theft he did not commit. The arrest was based on a facial recognition match that had misidentified him. Williams was held in jail for 30 hours before being released. The facial recognition system had matched his driver's license photo to surveillance footage of a different person.

When the detective showed him the surveillance image, Williams held it next to his own face and said: "I hope you don't think all Black people look alike." Williams was not the only case.

In 2019, Nijeer Parks, a Black man in New Jersey, was arrested for shoplifting and attempting to hit a police officer with a car — at a location he had never been to, based entirely on a facial recognition match. He spent ten days in jail and nearly a year fighting the charges before they were dismissed. In every documented case of a false facial recognition arrest in the United States, the person wrongly arrested has been Black. Every single one. This is not coincidence.

It is the predictable consequence of deploying a system that is forty times more likely to misidentify dark-skinned faces in a context where misidentification leads to arrest, and arrest falls on the person the algorithm points to.

The Keeper in the Clinic

In the courtroom, the keeper is the judge. In the clinic, the keeper is the doctor. A doctor who examines a patient, takes a history, runs tests, and makes a diagnosis is exercising judgment. The doctor can see things the algorithm cannot: the patient who minimizes their pain, the patient whose chart understates their condition because they have not been able to afford regular visits, the patient whose numbers look acceptable but whose face looks wrong. A doctor who knows their patient population — who understands the barriers to care that their patients face — can adjust.

A doctor can say: the score says this patient is low risk, but I have seen enough patients from this community to know that low cost does not mean low need.

But the doctor, like the judge, faces the structural pressure of the number. The algorithm has been validated. It has been endorsed by the institution. It processes 200 million patients. The doctor who overrides the algorithm must justify the override. The doctor who follows the algorithm needs no justification. The path of least resistance is deference, and deference to a biased output reproduces the bias — at scale, automatically, invisibly. The deskilled pilot has become the deferential doctor. The autopilot degrades the pilot's skill.

COMPAS degrades the judge's authority. The healthcare algorithm degrades the doctor's clinical judgment. In each case, the tool that was designed to assist the keeper becomes the mechanism by which the keeper's judgment is displaced.

The keeper remains at the bedside.

The keeper's role shrinks.

But the keeper who opened this chapter did not shrink. Dr. Sikaneta looked at the same data the algorithm would have used and asked whether the categories were adequate. He disaggregated when the system would have aggregated. He found the birth-country specificity that no standard classification captures. He demonstrated that the risk persists across decades in a universal system — a finding that changes clinical practice for the communities he serves. That is what it looks like when the keeper's role does not shrink. It is also, in the current trajectory of healthcare AI, the exception rather than the rule.

The trajectory has a date on it now. By early 2026, AI clinical decision support was no longer arriving — it had arrived, embedded in the workflows of hospitals and clinics, recommending diagnoses, flagging risks, prioritizing cases. In January 2026 the Food and Drug Administration loosened its oversight of this software, declining to regulate tools whose single recommendation a clinician can, in principle, independently review. The logic is the keeper's logic exactly: the doctor remains in the loop, free to look beneath the output. But the guidance places the liability on the

physician while the workflow places the pressure on the clock, and influence is harder to regulate than responsibility. The system keeps the doctor present and quietly rewards the doctor who does not argue with it.

The Compounding

There is a feature of algorithmic bias in healthcare that does not have an equivalent in the courtroom, and it is worth naming because it is the mechanism by which the harm deepens over time.

When the healthcare algorithm classifies a Black patient as low risk and denies them entry to the care management program, the patient does not receive the additional care.

Because the patient does not receive the additional care, the patient's health may deteriorate.

But the deterioration does not necessarily increase the patient's healthcare spending — the patient may still face the same barriers to care that kept their spending low in the first place. The patient gets sicker. The spending stays low. The algorithm, in the next cycle, still classifies the patient as low risk. The bias compounds. The algorithm's output — which denies care — produces the data — low spending — that the algorithm uses to justify denying care in the next cycle.

The system generates the evidence for its own correctness. A patient who is denied care looks, in the data, like a patient who does not need care. The denial becomes invisible. It looks like health. This is a feedback loop, and it is one of the most dangerous features of algorithmic systems deployed in consequential domains. The algorithm does not merely reflect the past. It bends the future toward its own assumptions. The criminal justice version of this loop is well documented: predictive policing algorithms direct more officers to neighborhoods flagged as high crime, which produces more arrests in those neighborhoods, which produces more data showing those neighborhoods are high crime, which justifies directing more officers there. The healthcare version is quieter but equally corrosive: the algorithm that denies care to the undertreated produces data showing the undertreated are low need, which justifies continuing to deny them care.

The Response That Worked

The Obermeyer paper is unusual in algorithmic bias research because it has a relatively hopeful coda. The researchers worked with the algorithm's developer, and

the company agreed to change the metric. Instead of predicting cost, the revised algorithm predicts a combination of health indicators that more directly measure need. The bias was reduced by approximately 84%. This matters because it demonstrates that the problem is not insoluble. The algorithm was not inherently racist. It was optimizing the wrong thing. Change the metric and the bias decreases.

This is evidence that careful, critical examination of algorithmic systems can identify and correct harms.

But the correction happened only because a team of researchers chose to investigate. For the years before the paper was published, the algorithm was operating as designed, endorsed by the institutions that used it, processing 200 million patients, and systematically underserving Black patients — and nobody inside the system detected the problem.

The keeper — in this case, the researchers who performed the audit — was not built into the system.

The keeper arrived from outside, years later, and only because academic researchers had the motivation, the access, and the statistical tools to look. There is a tension here that this chapter must name. The algorithm that encoded the bias was also the instrument that made the bias visible, measurable, and correctable — a function the human system had failed to perform for decades. Before the algorithm, human gatekeepers in the healthcare system were also failing Black patients — that is why spending was lower in the first place.

The disparity was real and the humans in the system did not detect it. The algorithm did not create the injustice. It inherited the injustice and, by encoding it consistently, made it examinable for the first time. The lesson is not that algorithms are biased. The lesson is that algorithms inherit the biases of the systems they learn from — and the correction requires a keeper who can recognize the bias once it is exposed. How many algorithms are running today, at similar scale, with similar biases, that have not yet been audited?

The answer is not known.

The question is the central policy question of algorithmic governance, and the answer is almost certainly: many.

The Architecture of Invisible Harm

The Therac-25 killed three patients. The families knew. The hospitals knew. The deaths were investigated, the machine was recalled, and the flaw was identified. The

harm was visible. The healthcare algorithm denied additional care to tens of thousands of Black patients over multiple years.

Nobody knew. No alarm sounded. No patient was told: you were denied this program because the algorithm scored you as low risk, and the algorithm scored you as low risk because you are Black and Black patients spend less on healthcare and the algorithm confused cost with need. The harm was invisible — not because it was small, but because it was distributed, statistical, and indistinguishable, at the individual level, from the ordinary workings of a healthcare system that already underserves Black patients. This is the architecture of invisible harm, and it is the architecture of the remaining chapters.

The sentencing machine produces bias that is invisible because it is dissolved into a number. The diagnostic produces bias that is invisible because it looks like health. The feed, in the next chapter, produces harm that is invisible because it looks like engagement — people clicking, sharing, commenting — while the content they are engaging with is driving them toward hatred and violence. The common thread: in each case, the system's own metrics say it is working. The risk score predicts cost accurately. The facial recognition system identifies most faces correctly.

The engagement metrics are up.

The system is succeeding by its own definition of success. The harm is in the gap between the system's definition of success and the human definition of success — and that gap is invisible to the system, invisible to its operators, and visible only to the people who fall into it.

“The system is succeeding by its own definition of success. The harm is in the gap between the system's definition of success and the human definition. That gap is visible only to the people who fall into it.”

The sentencing machine misread futures. The diagnostic misread the present. The next chapter tells the story of a system that did not misread anything at all. It read the data perfectly, optimized its objective function precisely, and achieved its goal exactly — and the goal, measured in clicks and shares and time on screen, was perfectly aligned with the amplification of content that drove a population toward genocide. The COMPAS algorithm was biased by accident. The healthcare algorithm was biased by proxy. The recommendation engine at the center of the next chapter was not biased at all.

It was optimized.

And the optimization produced mass atrocity.

CHAPTER TLK-9

The Feed

On Facebook's role in the Myanmar genocide, the recommendation engine, and the speed-judgment tradeoff

The Feed

On the Facebook recommendation engine, genocide, and the discovery that engagement optimization is indistinguishable from radicalization.

The previous two chapters told stories about algorithms that encoded bias by accident. COMPAS absorbed the racial disparities in the criminal justice system through its training data. The healthcare algorithm absorbed the racial disparities in healthcare access through its choice of metric. In both cases, the designers did not intend the harm. The harm was a side effect of a reasonable design choice interacting with an unequal world. This chapter tells a different story. The algorithm at the center of this chapter was not biased.

It was not making an error. It was not confused about its metric. It was optimizing, with extraordinary effectiveness, for the objective it was given: engagement. The problem is that engagement, measured at the scale of billions of interactions per day, is a metric that does not distinguish between a user clicking on a friend's birthday photo and a user clicking on content that tells them their neighbors are subhuman and should be exterminated. Both are engagement. Both generate the signal the algorithm is trained to maximize.

And the content that provokes the most intense emotional reaction — which is, reliably, content that divides, enrages, and dehumanizes — generates more engagement than the content that informs, connects, or entertains. COMPAS was biased by accident. The healthcare algorithm was biased by proxy. The recommendation engine was not biased at all. It was optimized.

And the optimization produced mass atrocity.

The Country

Myanmar is a country of approximately 54 million people in Southeast Asia, bordered by India, China, Bangladesh, Laos, and Thailand. For decades, it was governed by a military junta that restricted press freedom, political opposition, and access to information technology.

When the country began opening politically and economically in the early 2010s, internet access expanded rapidly — from roughly 1% of the population in 2010 to over 30% by 2016, driven almost entirely by inexpensive smartphones and mobile data plans. For most people in Myanmar, Facebook was the internet. This is not a metaphor. Mobile carriers sold data plans that included free or discounted access to Facebook. Many users did not distinguish between Facebook and the internet itself. The platform was where people read news, communicated with family, conducted business, and encountered information about the world beyond their immediate community.

Facebook had approximately 20 million users in Myanmar by 2018 — roughly one-third of the country's population. The Rohingya are a Muslim ethnic minority living predominantly in Rakhine State, in Myanmar's west. They have been subject to systematic discrimination by the Buddhist-majority state for decades — denied citizenship under a 1982 law, restricted in movement, education, and marriage, and subjected to periodic waves of communal violence. The tensions between the Rohingya and the ethnic Rakhine Buddhist population are deep, historical, and complex.

What happened on Facebook was not complex. It was an amplification machine pointed at a population primed for violence, with no one watching the output.

“For most people in Myanmar, Facebook was the internet. The platform was where a nation encountered the world. And the platform's algorithm was optimizing for engagement.”

The Content

Beginning in the mid-2010s, anti-Rohingya content proliferated on Facebook in Myanmar. Nationalist Buddhist monks, military-linked accounts, and coordinated networks of fake profiles posted and shared content that dehumanized the Rohingya: fabricated stories of Rohingya violence against Buddhists, manipulated images purporting to show Muslim atrocities, calls for the expulsion or elimination of the

Rohingya population, and inflammatory rhetoric that described the Rohingya as animals, invaders, and threats to Buddhist identity. The content was in Burmese. Facebook, at the time, had approximately two dozen content moderators who spoke Burmese — for a platform serving 20 million users.

The moderation was not merely inadequate. It was, at the scale required, impossible. A content moderation team of two dozen people cannot review the output of 20 million users producing content in a language the company’s automated systems did not reliably understand. The content was, in effect, unmoderated.

But it was not unamplified. Facebook’s recommendation algorithm did not create the anti-Rohingya content.

But it distributed it. The algorithm observed that users who saw inflammatory anti-Rohingya posts engaged with them at high rates — clicking, commenting, sharing, reacting. The algorithm interpreted this engagement as a signal of user interest and showed more similar content to those users and to users with similar profiles. The content that generated the most engagement was, predictably, the content with the strongest emotional charge: the most extreme, the most frightening, the most dehumanizing. The healthcare algorithm’s feedback loop operates here at planetary scale.

The algorithm shows inflammatory content. Users engage with inflammatory content. The algorithm sees the engagement and shows more inflammatory content. Users engage more. The cycle accelerates. The feed becomes progressively more extreme — not because anyone decided to radicalize the users, but because radicalization is what engagement optimization looks like when the content involves an ethnic minority that a significant portion of the population already distrusts.

The Violence

In August 2017, the Myanmar military launched what it called “clearance operations” in Rakhine State. Over the following months, security forces and allied militias burned Rohingya villages, committed mass killings, and carried out systematic sexual violence. Approximately 700,000 Rohingya fled across the border to Bangladesh, creating one of the largest refugee crises in modern history. The UN High Commissioner for Human Rights described the operations as “a textbook example of ethnic cleansing.” Multiple international bodies, including the International Court of Justice, have since characterized the violence as genocide.

In 2018, the UN Human Rights Council’s Independent International Fact-Finding Mission on Myanmar published its report. The mission had been established to investigate the human rights situation in Myanmar, with particular focus on the Rohingya crisis. The report’s findings regarding Facebook were unambiguous. The mission found that Facebook had been used to “spread hate, incite violence, and coordinate attacks” against the Rohingya. The chairman of the fact-finding mission, Marzuki Darusman, stated that Facebook had played a “determining role” in the crisis. The report documented specific instances in which content posted on Facebook — fabricated stories of Rohingya attacks, calls to violence by military-linked accounts, dehumanizing rhetoric from nationalist organizations — had preceded and accompanied specific acts of physical violence.

The Optimization

It would be comforting to describe Facebook’s role in the Myanmar genocide as a failure. A bug. A system that malfunctioned.

But that description is not accurate. The recommendation algorithm worked. It identified content that users engaged with. It showed users more content like it. It increased time on platform, increased interactions, increased the metrics that Facebook’s business model depends on. By every measure internal to the system, the algorithm was succeeding. The problem was not that the system failed to optimize. The problem was what optimization means when the objective is engagement and the context is a society on the verge of mass violence. Engagement optimization does not distinguish between engagement driven by curiosity and engagement driven by fear.

It does not distinguish between a user who clicks because they want to learn and a user who clicks because they are being radicalized. It does not distinguish between content that builds community and content that incites genocide. All of these produce clicks. All of these produce shares. All of these produce comments. All of these are, to the algorithm, the same signal. Every chapter before this one converges here: the danger of an automated system is not always that it fails.

Sometimes the danger is that it succeeds at the wrong objective. The Therac-25 failed to check the beam configuration. The autopilot failed and handed off to deskilled pilots. MCAS succeeded at pushing the nose down but the sensor was wrong. COMPAS succeeded at predicting cost but cost was a proxy for race. The healthcare algorithm succeeded at identifying high-spending patients but spending was a proxy for access.

And the recommendation engine succeeded at maximizing engagement but engagement was a proxy for outrage.

The Speed

The Myanmar case also demonstrates Pattern 4 — the Speed-Judgment Tradeoff — at a scale that dwarfs everything before it. Facebook processes billions of content interactions per day across its platforms. The recommendation algorithm operates continuously, in real time, evaluating content, predicting engagement, and distributing posts to feeds at a rate that no human moderation system can match. The two dozen Burmese-speaking moderators were not inadequate because they were insufficient in number — though they were. They were inadequate because the task itself is impossible at the speed the algorithm operates.

By the time a human moderator can evaluate a piece of content, the algorithm has already shown it to thousands or millions of users. The content has already generated engagement. It has already been shared. It has already been absorbed. In the Therac-25, the operator had seconds. In Air France 447, the pilots had three minutes. In the 737 MAX, the pilots had eleven minutes. On Facebook, the window for human intervention is measured in milliseconds — the time between the algorithm’s ranking decision and the content’s appearance on a user’s screen.

The keeper cannot operate at this speed. No human can.

The system has accelerated past the point where human judgment can intervene before the consequence. The Speed-Judgment Tradeoff reaches its endpoint here: a system so fast that inserting a human into the loop would require slowing the system to a pace incompatible with its function. Facebook’s recommendation algorithm works because it is instant. Making it slow enough for human review would make it something other than what it is. The speed is not a feature of the system. The speed is the system.

The Warnings That Were Ignored

Facebook was warned. This is documented. As early as 2013, civil society organizations in Myanmar raised concerns about the spread of anti-Muslim hate speech on the platform. Researchers, journalists, and human rights groups contacted Facebook repeatedly, identified specific accounts and networks spreading inflammatory content, and requested action.

In 2014, a group of technology researchers published a detailed report on the role of Facebook in amplifying communal tensions in Myanmar. Facebook’s response was slow, inadequate, and, for the most consequential period, effectively absent. The company did not invest in Burmese-language content moderation at a scale remotely proportional to the platform’s influence in the country. It did not prioritize the development of automated tools capable of detecting hate speech in Burmese. It did not establish local partnerships sufficient to monitor the platform’s effects on the ground.

The warnings were received. The warnings were not acted upon with the urgency the situation required.

In 2018, after the genocide was already underway, Facebook’s own internal assessment acknowledged that the company had not done enough. Mark Zuckerberg told Congress that the situation in Myanmar was “a terrible tragedy” and that the company was “too slow to act.” In a subsequent interview, he acknowledged that Facebook needed to “do more” but resisted the characterization that the platform bore significant responsibility for the violence.

The pattern is, by now, familiar to the reader. AECL was told the Therac-25 was harming patients and could not reproduce the error. Boeing was told by its own test pilots that MCAS was problematic and the schedule held. Facebook was told that its platform was being used to incite violence against an ethnic minority and did not invest in the moderation required to address it. In every case, the institution received the information and the institution did not act with the urgency the information demanded.

The institution’s confidence in its system — its engineering, its product, its platform — exceeded its willingness to believe that the system could produce catastrophic harm.

The Question of the Keeper

Where is the keeper in a recommendation engine? In the Therac-25, the keeper was a hardware interlock. In the cockpit, the keeper was the pilot. In the courtroom, the keeper was the judge. In the clinic, the keeper was the doctor. In each case, the keeper was identifiable — a specific mechanism or a specific person positioned at the point where the system’s output meets the human who is affected by it. A recommendation engine has no such point. There is no single moment where a human intervenes between the algorithm’s decision and the user’s exposure to the content.

There are billions of such moments, every day, and each one is automated.

The keeper would need to be present at every feed refresh, every recommendation, every ranking decision — and would need to evaluate each one for its potential contribution to radicalization, communal violence, or genocide in a specific cultural context. No human can perform this task. Neither can a thousand humans.

The system has scaled past the point where the keeper model, as defined in this book's opening chapter, can function.

What happened instead is worse than the keeper's absence. The burden was silently transferred.

Because the platform outpaced its human moderators — fewer than two dozen Burmese-speaking reviewers for twenty million users — the responsibility for distinguishing truth from incitement was offloaded onto the end user. Farmers in rural Myanmar, many of whom had only recently acquired smartphones and for whom Facebook was the internet, were implicitly asked to be the keepers of their own reality: to evaluate algorithmic recommendations for genocidal content in real time, without training, without context, and without any indication that what they were seeing had been selected by a machine optimising for engagement rather than accuracy.

Human psychology is not equipped for this task. We are pattern-matching animals wired to trust the information our environment presents to us. The algorithm exploited exactly that wiring.

The keeper was not removed.

The keeper was made into the victim — given responsibility without capacity, accountability without tools, and a feed that looked like the world but was, in fact, a machine's guess at what would keep them scrolling.

And that is precisely the point. The book began with a lighthouse keeper — one person, one light, one stretch of coast. Each subsequent chapter has expanded the scope: one treatment room, one cockpit, one courtroom, one clinic. The recommendation engine serves two billion users.

The question is no longer “where is the keeper?” The question is whether the concept of a keeper has meaning at a scale where no human judgment can touch the system's output before it reaches the people it affects.

The Diffusion of Accountability

Pattern 5 — the Accountability Gap — reaches its most extreme expression in this chapter.

When the Therac-25 killed patients, AECL was identifiable as the responsible party.

When the 737 MAX crashed, Boeing was indicted. Responsibility, however imperfectly assigned, could be traced. Who is responsible for the Myanmar genocide? The military commanders who ordered the operations bear primary responsibility. The soldiers who carried out the killings bear responsibility. The political leaders who enabled the military bear responsibility.

But what of the platform that amplified the content that dehumanized the victims?

What of the algorithm that ranked that content for maximum visibility?

What of the company that designed the algorithm, deployed it in a country it did not understand, ignored warnings that the platform was being used as a weapon, and did not invest in the moderation that might have blunted the amplification? Facebook did not order the genocide. Facebook did not intend the genocide. Facebook's algorithm did not know what the Rohingya were, what genocide is, or what the content it was amplifying would lead to. The algorithm knew that certain content generated more engagement. It showed more of that content.

The content happened to be calls for ethnic cleansing. The accountability gap here goes beyond blame diffused across a complex system, as in the 737 MAX. Algorithmic amplification is a new kind of causation, and existing legal and moral frameworks have no adequate vocabulary for it. Facebook is no author of the content, no publisher in the traditional sense, no broadcaster. It is a distribution system that amplifies content based on predicted engagement, without evaluating the content's truth, its social impact, or its potential to contribute to violence.

There is no precedent, in law or in ethics, for this kind of entity — and the absence of precedent is itself a form of impunity.

The Algorithmic Judge: A Reckoning

The last three chapters have traced a progression. In the courtroom, the algorithm encoded bias by accident — absorbing the racial disparities of the criminal justice system through its training data and producing risk scores that were used to keep human beings in cages. The harm was in the bias.

The keeper — the judge — was present but deauthorized. In the clinic, the algorithm encoded bias by proxy — using cost as a stand-in for need in a country

where cost and need diverge along racial lines. The harm was in the metric.

The keeper — the doctor — was present but displaced. In this chapter, the algorithm was not biased at all. It optimized its objective function with precision. The harm was in the objective itself — engagement, measured without reference to truth, safety, or human consequence.

The keeper — the content moderator — was absent at scale. Accident. Proxy. Optimization. Each step is harder to detect and harder to fight than the last. Accidental bias can be identified through audit and corrected through redesign. Proxy bias can be identified through careful analysis of the metric and corrected by choosing a better one.

But optimization bias — a system that harms people precisely *because* it is achieving its goal — cannot be corrected by making the system work better. It can only be corrected by changing the goal.

And changing the goal of a system that generates hundreds of billions of dollars in revenue requires a force external to the system: regulation, public pressure, or the kind of institutional conscience that the next chapters will test.

“Accidental bias can be audited and fixed. Proxy bias can be identified and the metric changed. But optimization bias — a system that harms because it succeeds — can only be corrected by changing the goal. And the goal generates hundreds of billions in revenue.”

From the Feed to the Battlefield

Part III asked what happens when algorithms judge. Part IV asks what happens when algorithms act — in the most consequential domains remaining: the selection of military targets, the surveillance of entire populations, and the engineering systems whose complexity exceeds human capacity to verify. The recommendation engine amplified content that contributed to genocide. The next chapter describes systems designed explicitly for the purpose that the recommendation engine served incidentally: identifying human targets. The AI targeting systems described in the next chapter do not amplify calls for violence.

They are the violence — automated, accelerated, and operating at a speed that reduces the human in the loop to a formality.

CHAPTER TLK-10

The Target

On AI-assisted military targeting, autonomous weapons, and the accountability gap at machine speed

The Target

On AI-assisted military targeting, and the moment when the human in the loop becomes a formality.

On July 3, 1988, the USS Vincennes, an American guided-missile cruiser operating in the Strait of Hormuz during the Iran–Iraq War, fired two surface-to-air missiles at an aircraft climbing away from Bandar Abbas International Airport. The aircraft was Iran Air Flight 655, a civilian Airbus A300 on a scheduled commercial route to Dubai. All 290 people aboard — including 66 children — were killed. The Vincennes was equipped with the Aegis Combat System, the most advanced naval warfare computer of its era. The Aegis system tracked the approaching aircraft, classified it, and presented data to the ship’s combat information center.

The crew, operating under the stress of an ongoing surface engagement with Iranian gunboats, interpreted the data as indicating a descending Iranian F-14 fighter on an attack profile. The aircraft was in fact ascending, on a standard commercial departure trajectory, broadcasting a civilian transponder signal. The subsequent investigation found that the Aegis system had correctly identified the aircraft’s altitude, speed, and transponder signal. The data was on the screens. The crew misread the data, or selectively attended to the data that confirmed their expectation of an attack, or were overwhelmed by the volume of information the system presented during a high-stress engagement.

The system provided the information.

The humans made the wrong decision. The Vincennes incident is the historical anchor for this chapter because it demonstrates, in 1988, the central tension that AI-assisted targeting amplifies thirty-five years later: a human presented with

machine-generated data, under time pressure, in conditions of stress, will tend to defer to the interpretation that the system's presentation makes most salient — even when the data, correctly read, contradicts that interpretation.

The keeper was in the loop.

The keeper had the information.

The keeper killed 290 civilians.

“The data was on the screens. The aircraft was ascending, broadcasting a civilian signal. The crew fired anyway. The keeper was in the loop. The keeper had the information. The keeper killed 290 civilians.”

The Acceleration

The Vincennes crew had approximately three minutes between initial detection and the decision to fire.

Three minutes to identify the aircraft, assess the threat, request authorization, and launch. In that window, the crew processed data from the Aegis system, communicated with higher command, and reached a conclusion that was wrong.

Three minutes was not enough time for the human judgment required. Modern AI-assisted targeting systems operate at a fundamentally different tempo.

The systems reported on by investigative journalists and described by military sources do not present data to a human and wait for a decision. They process vast quantities of surveillance data — signals intelligence, communications intercepts, movement patterns, social network analysis, geolocation data — and produce targeting recommendations. Not raw data for a human to interpret. Recommendations. Ranked lists. Predicted identities. Confidence scores. The shift from data presentation to recommendation is the shift that transforms the human's role.

When the Aegis system showed the Vincennes crew a radar track, the crew had to interpret it: is this track a threat? The interpretation was wrong, but it was a human act of judgment.

When an AI targeting system generates a list of recommended targets with confidence scores, the human's role shrinks from interpretation to approval.

The machine has already judged.

The human is asked to ratify the machine's judgment. The judge who ratified a risk score has become the officer who ratifies a target. The judge saw a score and was structurally pressured to defer. The military operator sees a recommendation and is structurally pressured to approve. The path of least resistance is deference — because

overriding the machine requires justification, and the machine has already done the analysis that the human would need to perform independently to reach a contrary conclusion.

The human does not have the time, the data access, or the analytical capacity to replicate the machine's work.

The human can approve or reject.

The human cannot meaningfully evaluate.

The Reports

In 2024, the Israeli-Palestinian investigative outlet +972 Magazine, in collaboration with the Hebrew-language publication Local Call, published a series of investigative reports based on interviews with Israeli intelligence officers describing AI-assisted targeting systems used during the military operations in Gaza. The reports described two systems in particular.

The first, referred to as “Gospel,” was described as a system that generates recommendations for military targets — buildings, infrastructure, and sites assessed to be linked to militant activity. Sources described Gospel as producing targeting recommendations at a rate that dramatically exceeded the capacity of previous intelligence processes. Where earlier methods might produce a limited number of vetted targets over weeks, Gospel was described as generating hundreds of targets in compressed timeframes.

The second, referred to as “Lavender,” was described as a system that identifies individuals assessed to be associated with militant organizations. Sources described Lavender as processing surveillance data to generate lists of individuals ranked by their predicted association with militant groups, with each individual assigned a numerical score. Sources stated that the system's output was treated, in practice, as equivalent to a targeting decision — that the human review of individual targets generated by Lavender was minimal, sometimes lasting as little as seconds per target.

Twenty seconds. To review a machine's recommendation to kill a human being. The reporting described a process in which the AI system performed the substantive analytical work — correlating data sources, assessing patterns of life, estimating association with militant networks — and the human performed a nominal review that one source compared to a formality.

The system generated the list.

The human approved the list. The strike was executed. Often, according to the reporting, strikes were carried out on targets while they were in their homes, where they were surrounded by family members, because targeting them in their homes was operationally simpler than targeting them in other locations.

The Rubber Stamp

The concept of a “human in the loop” is central to the international legal and ethical framework governing the use of lethal force. International humanitarian law requires that attacks distinguish between combatants and civilians, that the anticipated military advantage be proportional to the expected civilian harm, and that precautions be taken to minimize civilian casualties. These obligations presuppose a decision-maker — a human who evaluates the target, assesses proportionality, considers the civilian environment, and makes a judgment about whether the strike is lawful. A human who has twenty seconds to review a machine-generated target is not performing this function.

Twenty seconds allows no independent assessment of the intelligence behind the recommendation, no evaluation of the civilian environment around the target, no meaningful consideration of proportionality. Twenty seconds is enough time to look at a name, see a confidence score, and approve. The automation paradox has no more lethal expression.

The system generates targets so efficiently that the human review process cannot keep pace without being compressed to the point of meaninglessness. The more targets the system produces, the less time the human has per target. The less time the human has, the less meaningful the review. The less meaningful the review, the more the system’s output becomes, in practice, the final decision.

The human is in the loop.

The human has not been removed.

But the human’s role has been compressed to the point where “in the loop” and “out of the loop” are functionally indistinguishable.

The keeper is present.

The keeper is not keeping.

The Collateral Arithmetic

The +972 reporting described another dimension of the AI targeting process: the management of expected civilian casualties. Sources described a system in which each

targeting recommendation was accompanied by an estimated number of civilians likely to be killed in the strike. This estimate was generated by the system based on data about the target's known location, the building's occupancy patterns, and the weapon selected for the strike. Sources described a threshold: a pre-authorized number of civilian casualties deemed acceptable for targets of different ranks or assessed importance.

For lower-ranking targets, the acceptable number was described as relatively low. For higher-ranking targets, the acceptable number was described as significantly higher.

The system generated the target.

The system estimated the civilian cost.

The human compared the estimate to the threshold.

If the number was below the threshold, the strike was approved. The arithmetic matters.

The system makes no binary strike-or-don't decision. It runs a calculation: is the expected number of dead civilians acceptable given the assessed importance of the target?

And the target's importance was assessed by the same AI system that generated the recommendation.

The machine determines who the target is.

The machine determines how important the target is.

The machine estimates how many civilians will die.

The human compares two numbers and authorizes. Every step in this chain that requires judgment — is this person a combatant, how important are they, how many civilian deaths are acceptable — has been delegated to the machine or reduced to a numerical threshold.

The human's judgment has been replaced by the human's arithmetic.

And the arithmetic is easier to perform than the judgment — which is precisely why the system is designed this way.

The Legal Void

International efforts to regulate autonomous weapons have been underway for over a decade. The Convention on Certain Conventional Weapons (CCW), a UN framework, has hosted discussions on lethal autonomous weapons systems since 2014. Multiple countries and civil society organizations have called for a binding international treaty.

As of this writing, no such treaty exists. The core difficulty is definitional.

What counts as an autonomous weapon? A fully autonomous system that selects and engages targets without any human involvement is one end of a spectrum. An AI system that generates targeting recommendations reviewed by a human for twenty seconds is somewhere in the middle. A precision-guided munition that tracks a designated target after launch is at the other end. The line between “autonomous” and “assisted” is a gradient, not a bright line, and the gradient is defined by the quality of human involvement — a quality that, as this chapter has argued, can be degraded to the point of meaninglessness while remaining technically present.

Nowhere is the accountability gap wider.

If a soldier fires a rifle and kills a civilian, the chain of responsibility is clear.

If an AI system generates a target, a human approves it in twenty seconds, a drone strikes a building, and civilians die, who is responsible?

The system designer who built the algorithm? The intelligence analyst who trained it? The operator who approved the target? The commander who set the acceptable casualty threshold? The political leadership that authorized the use of the system? The answer, in practice, has been: all of them, and therefore none of them. The responsibility diffuses.

The system absorbs it.

And the civilians who die in the strike are recorded as collateral damage within acceptable parameters — parameters set by the same system that generated the target.

The Red Lines

In February 2026, an event occurred that connected this chapter directly to the thesis of this book. Anthropic, the company that builds the Claude AI system, was offered a contract with the US Department of Defense valued at approximately \$200 million. The company’s leadership evaluated the contract and identified two applications that it determined were incompatible with its values: the use of its AI for autonomous weapons targeting and for mass surveillance of civilian populations. Anthropic declined those specific applications while remaining open to other defense work.

The response was immediate and severe. Elements within the defense establishment pushed to designate Anthropic as a supply-chain risk — a classification that would effectively blacklist the company from government contracts. The argument was that a company unwilling to provide unrestricted AI capabilities to the military could not be relied upon as a strategic partner. The public responded

differently.

Within weeks of the confrontation becoming public, Claude became the number one application in the App Store. The company reported over one million daily sign-ups. The growth was driven predominantly by users who had never previously used the product — people who chose Anthropic specifically because the company had drawn a line. It deserves to be called what it is: the keeper's last stand in the AI era. Anthropic did not argue that AI should not be used by the military. It did not refuse all defense work.

It drew two specific lines: no autonomous weapons, no mass surveillance. These are the two applications where the keeper is most obviously removed from the loop — where the machine selects targets without meaningful human judgment, and where the machine monitors populations without individualized human review. The company that builds the AI said: these two uses remove the human from loops where the human must remain. Whether this line holds is an open question. The economic and political pressures to cross it are enormous. The precedent, as this entire book has documented, is that lines drawn against the removal of the keeper are eventually crossed — because the machine is faster, because the machine is cheaper, because the machine is more consistent, and because the people who benefit from the machine's efficiency do not bear the cost of the machine's errors.

The Pattern at Its Endpoint

This chapter is the convergence point of the book's five patterns.

The Reliability Trap: the more effective AI targeting becomes, the more dependent military operations become on it, the less capable human analysts become of performing the targeting function independently, and the worse the consequences when the system is wrong.

The Single Point of Trust: the system's targeting recommendation is treated as equivalent to a verified intelligence assessment.

One system.

One score.

One decision.

The Invisible Bias: the system is trained on data from previous conflicts, previous intelligence assessments, previous targeting decisions — all of which contain the biases and errors of those prior operations.

The patterns the system learns include the patterns of who was targeted before, which may not correspond to who should be targeted now.

The Speed-Judgment Tradeoff: the system generates targets faster than humans can review them.

The human review is compressed to the point of formality. The speed of the system is the mechanism that removes the human from the loop while maintaining the appearance of human control.

The Accountability Gap: when a strike kills civilians, responsibility diffuses across the system — the algorithm, the operator, the commander, the threshold, the policy.

Nobody is responsible because everyone is responsible.

The system absorbs the accountability that should attach to a human decision to take a human life.

The Doctrine

As this book was being completed, the targeting systems described in this chapter stopped being experimental. On February 28, 2026, the United States and Israel launched Operation Epic Fury — a massive joint military campaign against Iran that struck over 1,000 targets within the first 24 hours. Artificial intelligence tools were deployed in active targeting and intelligence support roles. The kinetic strikes were accompanied by AI-enabled cyber warfare, including the hacking of a widely used religious calendar app to deliver targeted psychological messaging to over five million users.

What was experimental in Gaza — Lavender, The Gospel, the AI targeting systems that generated lists of human beings for lethal action — had become standard operating doctrine for a major superpower conflict in less than two years. I want to name something here that the operational language of military planning is designed to obscure.

When a targeting system calculates that a strike on a building will kill twelve civilians and three combatants, and the operator approves the strike based on that calculation, the twelve civilian deaths are not incidental. They are not the fog of war. They are not imperfect intelligence. They were computed in advance.

The system modeled their deaths before they happened, assigned their lives a numerical weight, measured that weight against a military objective, found the weight acceptable, and proceeded. Those twelve people were dead in the model before the missile launched. That is not collateral damage. That is planned civilian killing with a

mathematical justification. International humanitarian law has always acknowledged that civilians die in war. The legal and ethical framework handles this through the doctrine of proportionality — the *unintended* civilian harm must not be excessive relative to the military advantage.

The key word is unintended.

But when a system computes the civilian deaths in advance, presents them to an operator as an acceptable ratio, and the operator approves — those deaths are not unintended. They are planned. They were modeled, calculated, weighed, and accepted before the weapon was fired. The mathematical justification is what makes it feel clean. A soldier who shoots a child experiences moral injury. A system that outputs “12 estimated civilian casualties, acceptable under current engagement parameters” experiences nothing. The moral weight evaporates into the calculation. The number creates distance between the decision and its consequence.

And the person who approves the number — in twenty seconds, as the reporting on Lavender documented — is not making a judgment. They are ratifying a computation.

The human in the loop is not exercising discernment. They are providing a signature.

“When the system models civilian deaths in advance, assigns their lives a numerical weight, and proceeds — those deaths are not collateral damage. They are planned. The number creates distance between the decision and its consequence, and the moral weight evaporates into the calculation.”

The speed of the transition — from experimental deployment to standard doctrine in less than two years — is itself the warning. The governance frameworks that might have constrained these systems were not built in time.

The window between “experimental” and “doctrine” closed before anyone outside the military could meaningfully intervene. That is the pattern this book has traced from its first page: the machine moves faster than the institutions charged with governing it.

From the Target to the Watcher

AI targeting selects individuals for lethal action. The next chapter describes the infrastructure that makes targeting possible — the surveillance systems that collect the data the targeting algorithms consume. Mass surveillance and autonomous weapons are one subject. Surveillance is the upstream infrastructure: the sensor network that

feeds the targeting system, the monitoring apparatus that identifies the individuals the algorithm will assess, and the mechanism by which entire populations are reduced to data points in a system optimized for a purpose those populations did not consent to and cannot escape.

CHAPTER TLK-11

The Watcher

On surveillance infrastructure, social credit systems, and the architecture of control

The Watcher

On mass surveillance infrastructure, the systems that monitor entire populations, and the keeper who watches everyone except itself.

In 2011, Malte Spitz, a German politician and privacy advocate, sued his mobile phone carrier, Deutsche Telekom, to obtain the records the company had collected about him under Germany's mandatory data retention law.

After a legal battle, Deutsche Telekom released the records. Spitz had them visualized by the German newspaper *Die Zeit*. The dataset covered six months. It contained 35,831 data points. Each point recorded Spitz's location, the time, and whether he was making a call, sending a text, or using mobile data. Plotted on a map and animated over time, the data produced a detailed reconstruction of Spitz's life: where he slept, where he worked, when he traveled, which routes he took, whom he visited, how long he stayed.

The visualization showed not just where Spitz was at any given moment but the pattern of his existence — the rhythms, the habits, the relationships made visible through proximity and frequency. That was 2011.

One person.

One carrier. Six months. 35,831 data points collected as a byproduct of normal phone use. The surveillance infrastructure described in this chapter does not collect data as a byproduct. It is purpose-built. It operates at the scale of entire populations.

And it is augmented by artificial intelligence that does not merely record data but interprets it — identifying faces, predicting behavior, flagging anomalies, and generating assessments of individuals who have not been accused of any crime, based on patterns the system has learned from data that the individuals did not consent to

provide.

“One person. One carrier. Six months. 35,831 data points. That was 2011.

The systems described in this chapter do not collect data as a byproduct.

They are purpose-built to monitor entire populations.”

The Laboratory

The most extensively documented mass surveillance system in the world operates in China’s Xinjiang Uyghur Autonomous Region. Since approximately 2017, the Chinese government has constructed an infrastructure of monitoring that encompasses virtually every dimension of daily life for the region’s approximately 12 million Uyghur inhabitants.

The system integrates multiple surveillance technologies into a unified monitoring platform. Facial recognition cameras are installed at checkpoints, intersections, mosques, markets, residential entrances, and along roads. Mobile phones are subject to mandatory installation of monitoring software that tracks communications, contacts, and content. DNA samples, voice recordings, and iris scans are collected from the population as part of a compulsory “health check” program. Wi-Fi probes at checkpoints harvest device identifiers. Movement between cities requires passing through security stations where identity documents are scanned and biometric data is verified.

The data from these sources feeds into an integrated platform that researchers and journalists have identified by several names, including the Integrated Joint Operations Platform (IJOP). The platform applies algorithmic analysis to the collected data and generates alerts when an individual’s behavior deviates from patterns the system considers normal. The deviations that trigger alerts include, according to leaked documents and research by Human Rights Watch, the Australian Strategic Policy Institute, and others: using a phone that was previously registered to another person, traveling to a neighboring country, having relatives abroad, praying more frequently than the system’s baseline, receiving a phone call from a foreign number, and using a VPN.

Individuals flagged by the system have been detained in a network of facilities that the Chinese government describes as vocational education and training centers and that international observers, governments, and the UN High Commissioner for Human Rights have characterized as arbitrary detention camps. Estimates of the number of people detained range from hundreds of thousands to over one million. The detentions

are based, in significant part, on algorithmic assessments generated by the surveillance platform.

The Weapon in the Pocket

Xinjiang's surveillance infrastructure is geographically bounded. A different surveillance technology operates without geographic limits, without the target's knowledge, and, in its most sophisticated form, without requiring the target to do anything at all. Pegasus is a surveillance tool developed by the Israeli company NSO Group. It is spyware — software designed to be installed on a target's mobile phone without their knowledge or consent. In its most advanced form, Pegasus can be installed through a “zero-click” exploit — an attack that requires no action by the target.

No link to click. No file to open. The phone is compromised through a vulnerability in the operating system or a messaging application, silently, without any indication to the user. Once installed, Pegasus provides the operator with complete access to the phone's contents and capabilities: messages, emails, photographs, contacts, location data, browsing history, calendar, passwords. It can activate the phone's microphone and camera without the user's knowledge, turning the device into a real-time surveillance instrument. The phone in the target's pocket becomes, without their awareness, a microphone in the room, a camera in their hand, and a complete record of their digital life.

The Citizen Lab at the University of Toronto, Amnesty International's Security Lab, and a consortium of media organizations have documented the deployment of Pegasus against journalists, human rights activists, lawyers, political opposition figures, and heads of state across dozens of countries. Targets have included journalists investigating government corruption in Mexico, human rights defenders in Saudi Arabia and the United Arab Emirates, opposition politicians in Rwanda and Poland, and associates of the murdered Saudi journalist Jamal Khashoggi — whose phone was reportedly compromised with Pegasus in the months before his assassination at the Saudi consulate in Istanbul in October 2018.

NSO Group maintains that Pegasus is sold exclusively to government agencies for the purpose of combating terrorism and serious crime. The documented deployments tell a different story: the technology is used, repeatedly and systematically, against the people who hold governments accountable — the journalists, the lawyers, the activists, the opposition politicians. The people who are, in the framework of this

book, the keepers of democratic accountability. The watcher is deployed against the keeper.

The Inversion

Until now, the keeper has been someone who stands between a system and the people the system affects. The hardware interlock between the beam and the patient. The pilot between the autopilot and the passengers. The judge between the algorithm and the defendant. The doctor between the risk score and the patient. Mass surveillance inverts this relationship.

The system does not remove the keeper from a process.

The system targets the keeper directly. Journalists who investigate state abuses are surveilled. Lawyers who defend political prisoners are surveilled. Activists who organize dissent are surveilled. The surveillance does not remove them from a loop. It undermines their capacity to function as keepers at all — by exposing their sources, mapping their networks, chilling their speech, and making them vulnerable to arrest, harassment, or worse. This is a qualitative shift in the book's argument. The Therac-25 removed a safety mechanism. The autopilot degraded a pilot's skill.

MCAS overrode the pilot's input. COMPAS deauthorized the judge. The recommendation engine outpaced the moderator. AI targeting compressed the reviewer to a rubber stamp. Each of these removed, degraded, or marginalized the keeper within a specific system. Mass surveillance attacks the keeper *as a category*. It does not degrade the keeper's function within one system. It degrades the keeper's ability to exist as a keeper in any system — by making the act of watching, questioning, and holding power accountable visible to the power being held accountable.

The keeper who knows they are watched is a keeper who self-censors, who avoids sensitive topics, who protects sources by not contacting them, who serves the function of accountability less effectively because the cost of serving it has been raised beyond what most people are willing to bear.

The Score

China's social credit system represents a different modality of surveillance — not the targeted monitoring of specific individuals but the continuous assessment of an entire population's behavior against a set of norms defined by the state. The social credit system, as implemented across various Chinese municipalities, aggregates data from

government records, financial transactions, court judgments, regulatory compliance databases, and, in some implementations, social media behavior and associations. Individuals and businesses receive scores or classifications that affect their access to services, their ability to travel, their eligibility for loans, and their public reputation.

Those classified as untrustworthy face restrictions: denial of air and rail tickets, exclusion from certain schools for their children, public listing on blacklists, and reduced access to government services.

The system is not a single, unified algorithm — it is a patchwork of local implementations with varying criteria, data sources, and consequences.

But the architecture is consistent: continuous data collection, algorithmic assessment of behavior against state-defined norms, and automated consequences for deviation.

The human decision-maker is not in the loop for individual assessments.

The system generates the classification.

The consequences flow automatically. The reader who has followed this book from the Therac-25 through COMPAS will recognize the architecture. A system that encodes the biases of the institution that built it (Invisible Bias). A system that operates faster than human review can match (Speed-Judgment Tradeoff). A system where responsibility for individual outcomes diffuses across the platform, the data sources, and the policy framework (Accountability Gap). A system whose reliability produces dependence that makes reversal difficult (Reliability Trap). A system whose assessments are based on data that is treated as ground truth without independent verification (Single Point of Trust).

All five patterns.

One system. Applied to 1.4 billion people.

The Export

The surveillance infrastructure developed in Xinjiang and tested in Chinese cities has not remained within China's borders. Chinese technology companies — including Huawei, ZTE, Hikvision, and Dahua — have exported surveillance systems to governments across Africa, Southeast Asia, Central Asia, Latin America, and the Middle East. These systems include facial recognition cameras, network monitoring tools, and integrated surveillance platforms based on architectures developed for domestic Chinese use. The Carnegie Endowment for International Peace documented in a 2019 study that AI surveillance technology had been deployed in at least 75

countries worldwide, with Chinese companies supplying technology to at least 63 of them.

The deployments include countries with documented records of authoritarian governance, suppression of political opposition, and human rights abuses.

The pattern is one this book has already named: the system is designed for one purpose (counterterrorism, public safety), deployed for that purpose in the initial context, and then repurposed for a different function (political surveillance, suppression of dissent) once the infrastructure is in place. The infrastructure does not distinguish between purposes. A facial recognition camera installed to identify criminal suspects works equally well to identify political protesters. A communications monitoring platform deployed to intercept terrorist messages intercepts all messages. The technology is dual-use not as an unfortunate side effect but as a structural property.

The same system that protects can oppress, and the difference is determined not by the technology but by the intent of the operator — an intent that can change with a new government, a new policy, or a new definition of who constitutes a threat.

The Democratic Watcher

It would be reassuring to frame mass surveillance as an authoritarian problem — a tool of dictatorships that democracies resist. The evidence does not support this framing. The Snowden disclosures of 2013 revealed that the US National Security Agency operated surveillance programs that collected the communications metadata of millions of American citizens without individual warrants. The UK's GCHQ operated programs that tapped undersea fiber-optic cables to intercept vast quantities of internet traffic. The "Five Eyes" intelligence alliance — the US, UK, Canada, Australia, and New Zealand — shared surveillance data across borders in ways that allowed each country to monitor activities that domestic law might prohibit it from monitoring directly.

Democratic governments deploy surveillance infrastructure for reasons that democratic publics often accept: counterterrorism, law enforcement, border security, pandemic response. The infrastructure, once deployed, persists. The legal authorities, once granted, expand. The surveillance capabilities developed for one crisis become available for the next, and the definition of what constitutes a crisis broadens over time. The difference between surveillance in a democracy and surveillance in an authoritarian state is not the technology. It is the institutional infrastructure that allows

the keeper to exist: courts that can review surveillance requests, legislatures that can limit authorities, press freedoms that protect the reporter who exposes abuses, and civil society organizations that can litigate against overreach.

These are not the keepers. They are the conditions under which keeping is possible. The journalist whose phone is hacked by Pegasus is the keeper — the individual human exercising judgment, asking questions, holding a light to the system. The court is what protects her. The legislature is what constrains the watcher. The press freedom is what allows the story to be published. Surveillance tools like Pegasus are used against all of them — not for what they run, but for what they protect: the infrastructure that keeps the keeper alive.

The watcher watches everyone.

But it watches the keeper most closely.

Because the keeper is the only threat the watcher cannot predict, cannot quantify, and cannot optimize away: the human who looks at the system itself and asks whether it should exist.

“The difference between surveillance in a democracy and in an authoritarian state is not the technology. It is the infrastructure that allows the keeper to exist — the courts, the legislatures, the press freedoms. Pegasus targets all of them. Not because they are the system, but because they are the conditions under which keeping is possible.”

The Pipeline

The previous chapter described AI systems that select targets for lethal action. This chapter describes the infrastructure that feeds those systems. The connection is operational, not metaphorical. Gospel, Lavender, and their equivalents in other militaries require data. The data comes from surveillance: signals intelligence, communications intercepts, location tracking, facial recognition, social network analysis. The more comprehensive the surveillance, the more data the targeting system has. The more data the targeting system has, the more targets it generates.

The more targets it generates, the less time the human has to review each one. Surveillance and targeting are not separate systems. They are stages in a pipeline. The watcher feeds the target. The target feeds the weapon.

The human in the loop — the keeper who is supposed to ensure that the pipeline produces lawful, proportionate, and discriminate outcomes — sits at the end of a process that has already determined the conclusion.

The keeper's twenty-second review is the final stage of a pipeline that began with a camera on a street corner, a phone in a pocket, or a metadata record on a server. This is why Anthropic's two red lines — no autonomous weapons and no mass surveillance — are not two separate positions. They are one position. The surveillance and the targeting are a single system with a single logic: identify the target and act on the identification. Drawing a line against one without drawing a line against the other is drawing no line at all.

From the Watcher to the Light

This chapter and the one before it have described the most consequential applications of automated systems: the selection of targets for killing and the monitoring of populations for control. In both cases, the keeper was not merely removed but targeted — the surveillance infrastructure described in this chapter is designed, in part, to identify and neutralize the people who might hold the system accountable.

The final chapter returns to where this book began: a lighthouse on a rock in the Atlantic, and a man who kept the light burning because he understood that the light was not for him.

The question is whether anyone still will.

CHAPTER TLK-12

The Light

On the return to the lighthouse, the five patterns, and the closing argument

The Light

The lighthouse at Smalls still stands. It is no longer the wooden structure on stilts that Henry Whiteside built in 1776, the one Thomas Howell clung to through Atlantic storms while keeping the light burning beside his dead companion. That lighthouse was replaced in 1861 by a stone tower, and the stone tower was automated in 1987. The last keepers left. The light now operates by electricity, monitored remotely from the mainland. It is more reliable than any human keeper could have maintained it. It does not get tired.

It does not get sick. It does not need food or rest or relief. The light shines every night. On average, it is better than the keeper it replaced.

This book has not been about the average.

“The light shines every night. On average, it is better than the keeper it replaced. This book has not been about the average.”

What the Book Found

This book began with a question: what happens when the person in the loop is removed? It traced that question through twelve chapters, across more than a century, from a lighthouse on a rock to an AI system that generates military targets.

What it found was not a collection of unrelated failures. It found a pattern — five patterns, in fact — that repeat with a consistency that would be remarkable if it were not also predictable.

The Reliability Trap ran through every domain. On the Titanic, the ship’s record produced the confidence that cut the lifeboats. In the cockpit of Air France 447, the

autopilot's dependability wore away the pilots' hands. In the courtroom, the algorithm's consistency displaced the judge's willingness to disagree. In military targeting, the system's efficiency compressed human review to twenty seconds. The trap is the same everywhere: the system's success is the mechanism of the human's decline.

The Single Point of Trust took different bodies: software standing in for hardware interlocks in the Therac-25, one sensor commanding a system that could overpower the pilots of the 737 MAX, a single algorithmic assessment treated as verified intelligence in AI targeting.

The pattern is the same: trust concentrated in a single system, without independent verification, is trust that will eventually be betrayed.

The Invisible Bias worked the same laundering everywhere: COMPAS washed racial discrimination through proxy variables, the healthcare algorithm confused cost with need, and facial recognition trained on lighter-skinned faces went selectively blind.

The pattern: the data reflects the world as it is, the algorithm learns the world as it is, and the output reproduces the world's inequities while appearing to transcend them.

The Speed-Judgment Tradeoff tightened with each chapter. The Therac operator had seconds. The Air France pilots had three minutes. Facebook's algorithm distributed content in milliseconds, and the targeting system now generates targets faster than humans can evaluate them. The window for human judgment is shrinking, and at sufficient speed the human is not in the loop but beside it. That trajectory poses a challenge this argument must confront directly.

If the system operates faster than human thought, what does "keeping the keeper" mean? The answer is not faster humans. The answer is systems that are not permitted to execute consequential decisions at speeds that preclude human review. The objection is obvious: the adversary who does not slow down gains the advantage. The response is equally uncomfortable: the advantage of speed without judgment is the advantage of a weapon without a safety. It will fire faster. It will also fire at the wrong target faster.

The keeper's function is not to match the machine's speed.

The keeper's function is to ensure that the machine does not exceed the speed of judgment.

The Accountability Gap widened as the systems grew. AECL could not reproduce the error and declared the machine safe. Boeing's responsibility dissolved across the FAA, the airline, and a sensor manufacturer. Facebook's amplification introduced a

causation for which existing frameworks have no vocabulary. In military targeting, the chain of responsibility dissolves into the system itself.

The pattern: the more automated the system, the more diffuse the accountability, and the less likely that anyone bears the cost of the failure.

The Arc

The book also traced an arc. In Part I the keeper was present and overruled — the Titanic’s officers had the warnings and trusted the ship. In Part II the keeper was removed, degraded, and overridden in sequence, each step leaving the human with less capacity to intervene. In Part III the keeper was deauthorized and outpaced: the judge given a number that discouraged independent judgment, the doctor given a metric that displaced clinical intuition, the moderator outrun by an algorithm no human could match. The system no longer removed the keeper; it made the keeper’s role nominal. In Part IV the keeper was rubber-stamped, targeted, and overwhelmed — twenty seconds per target, and a surveillance apparatus aimed at exactly the people whose function is to hold systems accountable.

And the complexity of modern software crossed a threshold beyond which human verification became not difficult but impossible.

The keeper has functioned, in the cases this book has documented, only when an individual with deep expertise and institutional independence chose to look beneath the number.

The question is not whether such keepers exist. They do.

The question is whether the systems being built will make room for them — or whether the algorithm will make them unnecessary before the next one arrives. The arc moves in one direction: from the keeper present and overruled, to the keeper absent and irrelevant. Every era has justified the movement in the same terms: the system is faster, cheaper, more reliable, more consistent, more objective. The justification is, on the merits, largely correct.

The machine *is* better than the human on average, in most conditions, most of the time. The book does not dispute this. The book disputes the conclusion drawn from it.

The Conclusion That Does Not Follow

The conclusion drawn, in every era and every domain this book has examined, is that because the machine is better on average, the human backup can be reduced. The lifeboats can be cut from 48 to 20. The hardware interlock can be replaced with

software. The pilot can be allowed to deskill. The judge can defer to the score. The reviewer can approve in twenty seconds. The content can be distributed without human review. The surveillance can operate without individual warrants. This conclusion does not follow.

The value of the keeper is not measured by average performance. It is measured by performance at the margin — at the moment when the system encounters conditions it was not designed for, when the data is wrong, when the assumption fails, when the metric is a proxy for the wrong thing.

The keeper's value is concentrated in the tail of the distribution: the 0.01% of events where the system is not merely imperfect but catastrophically wrong.

And because the cost of catastrophic failure — in lives, in liberty, in justice — is orders of magnitude larger than the cost of maintaining the keeper, the economic calculation that justifies removing the keeper is a calculation that discounts the catastrophe. It discounts the catastrophe because the catastrophe is rare. It is rare because the system is reliable.

The system is reliable because it was well-designed.

And the conclusion is: because the system is well-designed, the keeper is unnecessary.

But the keeper is what makes the system survivable when the design fails.

And the design always, eventually, fails. Not on most nights. Not on the average night. On one night. The night the ice crystals block the pitot tubes. The night the sensor gives the wrong reading. The night the code encounters a value it was not written for. The night the algorithm confuses cost with need. The night the engagement metric amplifies a call for genocide. The night the confidence score is wrong and the strike kills a family. On that night, the keeper is the only thing standing between the system's failure and its consequence.

And in every case the evidence shows: on that night, the keeper was not there.

What the Keeper Must Become

This book does not argue against automation. The autopilot saves more lives than it costs. The algorithm processes more data than any human can. The surveillance system identifies genuine threats that human intelligence misses. The argument is not that machines should be removed. The argument is that the keeper must remain.

But the keeper of the twenty-first century cannot be the keeper of the nineteenth. Thomas Howell kept one light on one rock. These systems serve billions of people

simultaneously, operate at speeds no human can match, and contain complexities no human can fully understand. The lighthouse keeper model — one person, one system, continuous watchfulness — does not scale.

The keeper must evolve.

The keeper must become an institution, not just an individual. It must become a set of structures: independent audit bodies with the authority and technical capacity to examine algorithmic systems before and after deployment. Mandatory disclosure requirements that make the behavior of consequential algorithms visible to the people they affect. Red teams empowered to find the failure before the failure finds the patient, the passenger, the defendant, the civilian. Kill switches that allow human override of automated systems in safety-critical domains, tested not just for existence but for physical executability — because a cutout switch that cannot be turned against the aerodynamic forces the system has created is not a safety mechanism but a fiction.

The keeper must also become a culture — an institutional commitment to the principle that the efficiency of automation does not justify the elimination of human judgment in domains where the cost of failure is measured in lives. Aviation learned this, imperfectly and at great cost. Medicine is learning it. Criminal justice is resisting it. Military operations are moving in the opposite direction.

And the technology industry, which builds the systems that all other domains depend on, has not yet decided.

The Company That Drew a Line

In February 2026, a company that builds AI systems was offered a contract to provide its technology for military applications. The company's leadership identified two specific uses that were incompatible with their assessment of where the human must remain in the loop: autonomous weapons targeting and mass surveillance of civilian populations. They said no to those two uses. The response from the defense establishment was pressure to designate the company as a supply-chain risk. The response from the public was different. Millions of people chose the company's product specifically because it had drawn a line.

The market rewarded the keeper. This is a data point, not a conclusion.

One company.

One decision.

One moment in which the economic incentive and the ethical position aligned. The documented history provides no basis for confidence that this alignment will hold.

The economic pressures to remove the keeper are persistent and structural. The political pressures are real. The institutional incentives favor efficiency over caution, speed over judgment, automation over oversight.

But it happened. A company that builds the technology said: not this use. Not this loop.

The human stays.

And the public, in numbers large enough to measure, said: yes. That is the company we trust. That is the product we choose.

The keeper's survival may depend less on regulation than on this: on the discovery that people, given the choice, prefer the system that keeps the human in the loop to the system that removes them. That the market for trust is larger than the market for efficiency. That the keeper is not a cost but a value — a thing people will pay for, choose, and defend. Whether this is enough remains to be seen.

The Deeper Question

This book has documented the cost of removing the keeper.

But it has not answered the question that the evidence inevitably raises: *why* does the removal keep happening?

The keeper was removed from the Therac-25 because the software was believed to be sufficient.

The keeper was degraded in the cockpit because the autopilot was more reliable.

The keeper was overridden in the 737 MAX because the schedule held.

The keeper was deauthorized in the courtroom because the algorithm was more consistent.

The keeper was outpaced on the platform because the engagement metric moved faster than any human could review. In every case, the reasoning was the same: the machine is better on average.

But behind that reasoning is a deeper pattern — a pattern that is not about individual machines or individual decisions but about the concentration of computational power itself. Every era of computing has produced a cycle: the capability distributes, and then the control reconcentrates. The mainframe concentrated power in a handful of companies. The personal computer appeared to distribute it. The internet appeared to distribute it further.

And then five companies recaptured the infrastructure, the data, and the distribution channels.

The keeper is removed not by accident but by the structural economics of each era's reconcentration — because the entities that accumulate power are the entities that benefit most from replacing human judgment with machine efficiency.

This book documented the cost. The deeper question — the five-thousand-year history of how computational power concentrates, why the pattern repeats, and whether the next era of computing will make the concentration permanent — is a question that demands its own investigation. It is the question that haunts the evidence this book has presented: not just what happens when the keeper is removed, but who benefits from the removal, and whether the accumulation of power that drives the removal can be governed before it becomes irreversible.

The Light

Thomas Howell kept the light burning through storms that would have killed a lesser person, beside a body that was decomposing in the salt air, on a rock twenty miles from shore with no relief and no communication with the mainland. He did this because the light was not for him. The light was for the ships. The ships could not see the rocks. The ships could not know the currents. The ships depended on a human being who sat in the dark, in the cold, in the storm, and kept the light burning because someone had to.

Not because any analysis had determined that the keeper's salary was justified by the expected reduction in shipwrecks.

Because without the light, the ships would break on the rocks.

And the keeper was the light. The light at Smalls still shines. The mechanism is reliable.

The keeper is gone.

And this book has spent twelve chapters documenting what happens, in domain after domain, era after era, when the keeper is gone and the mechanism encounters a night it was not designed for. The nights will come. They always do. Not one night — that framing is too comfortable, too contained, too much like a story with an ending. The failures will keep arriving for as long as the systems operate, because no system is perfect and the world is not static. The ice crystals will form in the pitot tubes.

The sensor will give the wrong reading. The algorithm will confuse cost with need. The engagement metric will amplify the call for violence. The targeting system will assign a high confidence score to the wrong person. The surveillance system will flag the journalist, the lawyer, the activist. The code will encounter a value it was not

written for. These are not hypothetical failures. Every one of them has already happened. Every one of them has already killed, harmed, imprisoned, or surveilled people who did not consent to the system that failed them.

The failures will continue. The only question left is the one the book began with — whether, when the next night comes, and it is already coming, there will be a keeper watching the light.

“The nights will come. They always do. The failures will keep arriving for as long as the systems operate. When the next night comes — and it is already coming — will there be a keeper watching the light?”

What to Read Next

This book asked what happens when the human is removed from the loop. If the evidence it presented left you with the question it left me with — *why does the removal keep happening?* — then the answer may lie in a history far older and far larger than the century of failures documented here. *Who Holds the Jar?* traces the concentration of computational power across five thousand years — from the shepherd’s clay counting jar in ancient Mesopotamia through every era of computing to the quantum and biological substrates that may define the next. It identifies a pattern that has repeated in every era: power distributes, then reconcentrates. Each cycle appears to democratize. Each cycle produces reconcentration. And the current cycle — AI, quantum computing, biological computation — may produce a concentration of power that, for the first time in history, cannot be reversed. If *The Last Keeper* is the human cost, *Who Holds the Jar?* is the reason the cost keeps being paid. It is available now as the second book in *The Cost of the Machine* series.

The Cost of the Machine Book One: The Last Keeper

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What happens when the human is removed from the loop?

Book Two: *Who Holds the Jar?*

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Who will control computational power — and will it be possible to take it back?

Book Three: The Great Convergence

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Who writes the rules — and what happens when nobody does?

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About the Author

Pumulo Sikaneta works in technology strategy, where he has spent years studying how organizations adopt, govern, and are transformed by the systems they build.

His professional work in partner enablement and solutions architecture has given him a close-up view of the gap between what technology promises and what it delivers — and of the humans who stand in that gap. It began as a question — why does the same pattern of failure repeat across every domain where machines replace human judgment? — and became a book when the evidence demanded more than a question.

The author's brother, Dr. Tabo Sikaneta, is a nephrologist at the Scarborough Health Network in Toronto and a founding member of the African Caribbean Kidney Association. His two decades of clinical work documenting the disproportionate burden of kidney disease in immigrant and non-White communities — including his 2025 BMJ Open study demonstrating that these disparities persist even within Canada's universal healthcare system — provided the human evidence that grounds this book's argument about medicine: that no algorithm can be more just than the society whose data it consumes.

The author was born in Zambia, raised in Canada, and lives in the United States. He believes that the most important skill in any field is the willingness to be corrected. Curious by nature. Correctable by choice. Always learning. Often wrong. Let's discuss.